

PART 1 : PHYSICS**SECTION-I**

1) A man can swim at a speed of 3 kmh^{-1} in still water. He wants to cross a 500 m wide river flowing at 2 kmh^{-1} . He keeps himself always at an angle of 120° with the river flow while swimming. A boy standing on the ground and a girl on the boat (drafting with the river velocity) observes the man. Then choose the correct option(s) :

- (A) The girl observes that the man is moving with a speed which is less than 3 km/hr.
- (B) The girl observes that the man is moving with a speed which is greater than 3 km/hr.
- (C) The boy observes that the man is moving with a speed which is less than 3 km/hr.
- (D) The girl observes that the man is moving with a speed which is equal to 4 km/hr.

2) A particle moves in a straight line, according to the law $x = 4a \left[t + a \sin \left(\frac{t}{a} \right) \right]$, where x is its position in meters, t in sec. and a is some constants, then the velocity is zero at :-

- (A) $x = 4a^2\pi$ meters
- (B) $t = \pi$ sec.
- (C) $t = 0$ sec
- (D) none

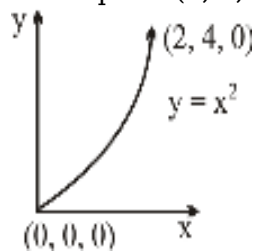
3) A wet ball is projected horizontally at speed of $u = 10 \text{ m/s}$ from the top of tower $h = 31.25 \text{ m}$ high. Water drop detach from the ball at regular time interval of $\Delta t = 1 \text{ sec}$ after throw. How many drops will detach from the ball before it hits the ground.

- (A) 2
- (B) 3
- (C) 4
- (D) 5

4) A block lying on a long horizontal conveyer belt moving at a constant velocity receives a velocity $v_0 = 5 \text{ m/s}$ relative to the ground in the direction opposite to the direction of motion of the conveyer belt. After $t = 4 \text{ s}$ the velocity of the block becomes equal to the velocity of the belt. The coefficient of friction between the block and the belt is 0.2 then what is the velocity (in m/s) of the conveyer belt?

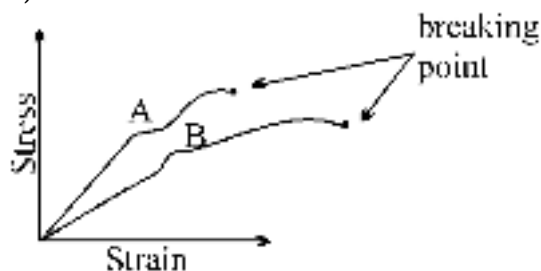
- (A) 4
- (B) 5
- (C) 3
- (D) 2

5) By applying a force $\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$ a particle is taken along path $y = x^2$ from point $(0, 0, 0)$ to the point $(2, 4, 0)$. The work done by the F on the particle is:



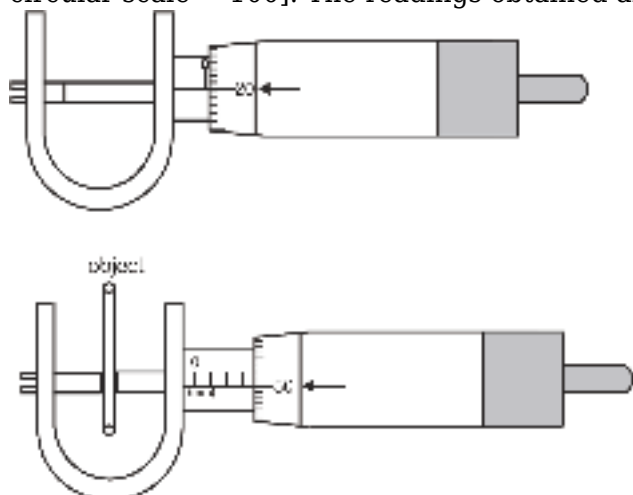
- (A) $\frac{280}{5}$
 (B) $\frac{140}{5}$
 (C) $\frac{232}{5}$
 (D) $\frac{192}{5}$

6) Select the correct statement on the basis of the given graph :-

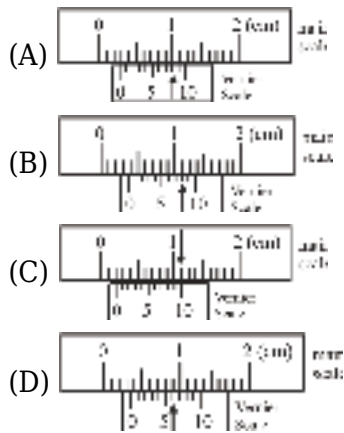


- (A) Young's modulus of A is greater but it is less ductile
 (B) Young's modulus of A is greater and it is more ductile
 (C) Young's modulus of A is less and it is less ductile
 (D) Young's modulus of A is less but it is more ductile

7) When a piece of wire is held diametrically in a screw gauge [pitch = 1mm, number of division on circular scale = 100]. The readings obtained are as shown:



Now if we measure the same piece of wire with help of vernier callipers [1 MSD = 1mm, 10 divisions of vernier coinciding with 9 divisions of main scale] having a negative zero error of 0.5 mm, then find which of the following figures correctly represents the reading shown.

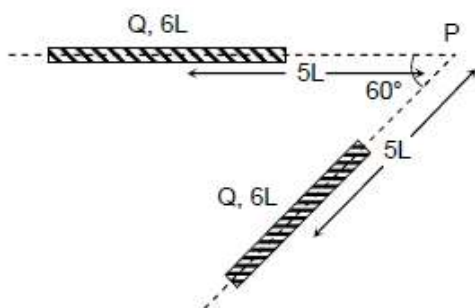


8)

In a Searle's experiment for determination of Young's Modulus, when a load of 50 kg is added to a 3 meter long wire micrometer screw having pitch 1 mm needs to be given a quarter turn in order to restore the horizontal position of spirit level. Young's modulus of the wire if its cross sectional area is 10^{-5} m^2 is :-

- (A) $6 \times 10^{11} \text{ N/m}^2$
 (B) $1.5 \times 10^{11} \text{ N/m}^2$
 (C) $3 \times 10^{11} \text{ N/m}^2$
 (D) None

9) Two rods each of length $6L$ carry charge Q are placed as shown in figure. Find magnitude of electric field at point P, which is at a distance of $5L$ from centre of each rod.

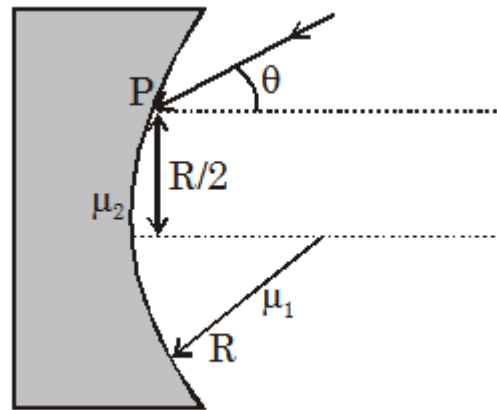


- (A) $\frac{1}{64\pi\epsilon_0} \cdot \frac{\sqrt{3}Q}{L^2}$
 (B) $\frac{\sqrt{5}Q}{4\pi\epsilon_0 L^2}$
 (C) $\frac{1}{2\pi\epsilon_0} \cdot \frac{Q}{L^2}$
 (D) $\frac{3Q}{5\pi\epsilon_0 L^2}$

10) A carbon dioxide laser emits a sinusoidal electromagnetic wave that travel in vacuum in the negative X-direction. The wavelength is $10.6 \mu\text{m}$ and the $\vec{E}$ field is parallel to +ve z-axis with $E_{\text{max}} = 1.5 \text{ Mv/m}$. Write vector equation of $\vec{B}$ as function of time and position.

- (A) $\vec{B}(x, t) = \hat{j}(5.0 \times 10^{-3} \text{T}) \cos[(5.93 \times 10^5 \text{rad/m})x + (1.78 \times 10^{14} \text{rad/s})t]$
- (B) $\vec{B}(x, t) = -\hat{j}(5.0 \times 10^{-3} \text{T}) \cos[(5.93 \times 10^5 \text{rad/m})x + (1.78 \times 10^{14} \text{rad/s})t]$
- (C) $\vec{B}(x, t) = \hat{k}(1.5 \times 10^6 \text{V/m}) \cos[(5.93 \times 10^5 \text{rad/m})x + (1.78 \times 10^{14} \text{rad/s})t]$
- (D) $\vec{B}(x, t) = -\hat{k}(1.5 \times 10^6 \text{V/m}) \cos[(5.93 \times 10^5 \text{rad/m})x + (1.78 \times 10^{14} \text{rad/s})t]$

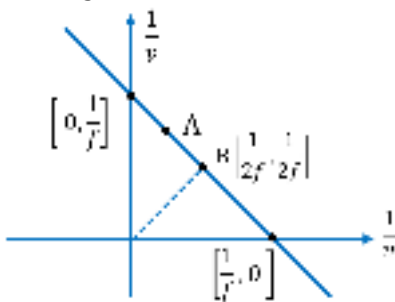
11) The minimum value of θ so that the ray gets totally reflected from spherical surface of radius of



curvature R is $\left[\frac{\mu_2}{\mu_1} = \frac{1}{2} \right]$

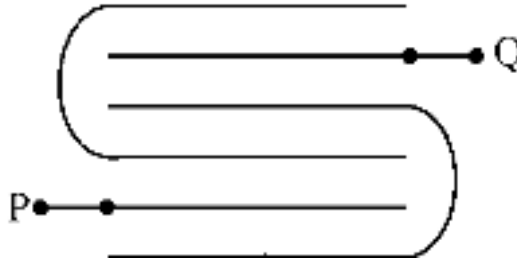
- (A) 30°
- (B) 60°
- (C) zero
- (D) not possible

12) Figure shows $\frac{1}{v}$ vs $\frac{1}{u}$ curve for convex mirror. Nature of image at point A is :-



- (A) Real, erect, magnified
- (B) virtual, inverted, magnified
- (C) virtual, inverted, diminished
- (D) virtual, erect, diminished

13) Identical metal plates are located in air at equal distances d from one another. The area of each plate is equal to A . If plates are interconnected as shown in figure, then the capacitance of system



between points P and Q is :-

- (A) $\frac{\epsilon_0 A}{2d}$
- (B) $\frac{\epsilon_0 A}{d}$
- (C) $\frac{2\epsilon_0 A}{d}$
- (D) $\frac{4\epsilon_0 A}{d}$

14) Two particles of combined mass M , placed in space with certain separation, are released. Interaction between the particles is only of gravitational in nature and there is no external force present. Acceleration of one particle with respect to the other when separation between them is R , has a magnitude :

- (A) $\frac{GM}{2R^2}$
- (B) $\frac{GM}{R^2}$
- (C) $\frac{2GM}{R^2}$
- (D) 0

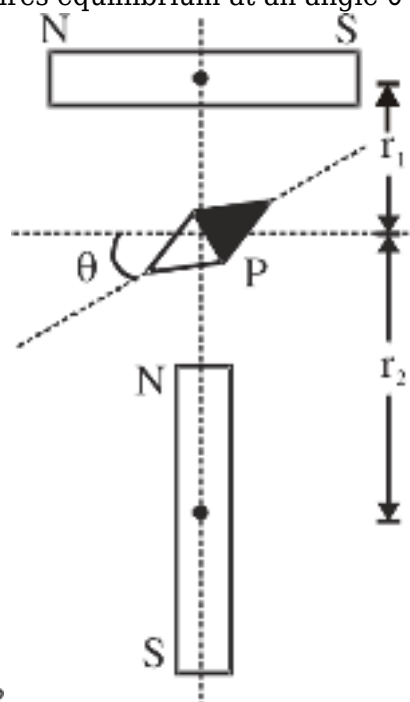
15) Two spherical bodies A (radius 6cm) and B (radius 18cm) are at temperatures T_1 and T_2 respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B at 1500 nm. Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B?

- (A) 9
- (B) 12
- (C) 16
- (D) 27

16) An ideal gas is in thermodynamic equilibrium. The number of degrees of freedom of a molecule of the gas is n . The internal energy of one mole of the gas is U_n and the speed of sound in the gas is v_n . At a fixed temperature and pressure, which of the following is the correct option ?

- (A) $v_3 < v_6$ and $U_3 > U_6$
- (B) $v_5 > v_3$ and $U_3 > U_5$
- (C) $v_5 > v_7$ and $U_5 < U_7$
- (D) $v_6 < v_7$ and $U_6 < U_7$

17) Two identical short bar magnets placed in horizontal plane according to figure. A small compass needle acquires equilibrium at an angle θ in the effect of both magnets at point 'P' then value of ratio



of r_1 and r_2

(A) $\left(\frac{2}{\tan \theta}\right)^{1/3}$

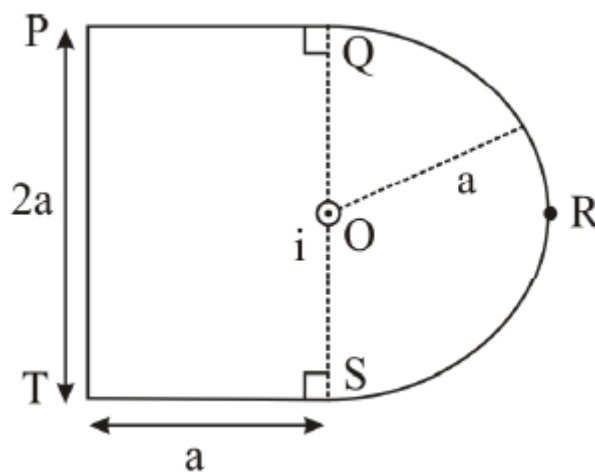
(B) $\left(\frac{\tan \theta}{2}\right)^{1/3}$

(C) $\left(\frac{\cot \theta}{2}\right)^{1/3}$

(D) $\left(\frac{2}{\cot \theta}\right)^{1/3}$

18) Consider an amperian loop PQRT (QS is diameter of semicircle QRS) around an infinitely long

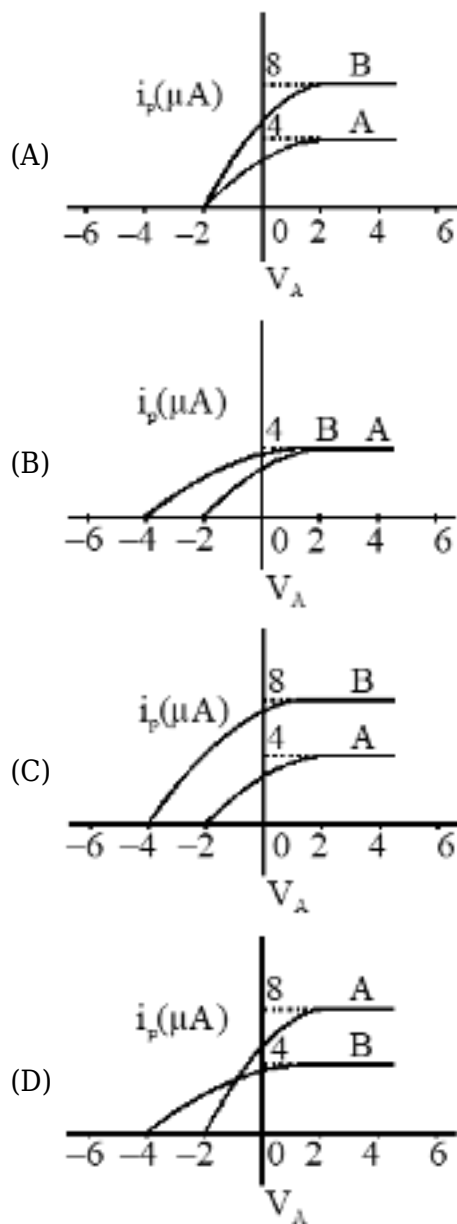
current carrying wire at O (current comes out of plane). The value of line integral $\oint \vec{B} \cdot d\vec{\ell}$ QPTS



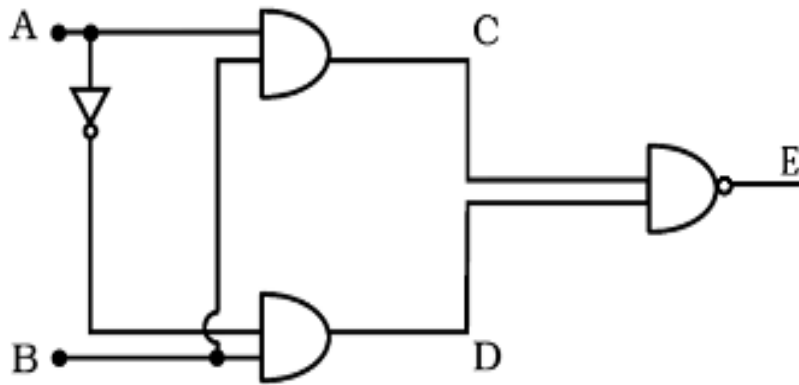
(traversed anti clockwise) is:

- (A) $\frac{\mu_0 i}{2}$
 (B) $> \frac{\mu_0 i}{2}$
 (C) $< \frac{\mu_0 i}{2}$
 (D) $\frac{4\mu_0 i}{\pi}$

19) In a photoelectric effect experiment, photons of energy 5 eV are incident on the photo-cathode of work function 3 eV. For photon intensity $I_A = 10^{15} \text{ m}^{-2} \text{ s}^{-1}$, saturation current of 4.0 mA is obtained. Sketch of the variation of photocurrent i_p against the anode voltage V_a for photon intensity I_A (curve A) and $I_B = 2 \times 10^{15} \text{ m}^{-2} \text{ s}^{-1}$ (curve B) will be :



20) Truth table for the given circuit (Fig) is :



(A)

A	B	E
0	0	0
0	1	0
1	0	0
1	1	1

(B)

A	B	E
0	0	1
0	1	1
1	0	1
1	1	1

(C)

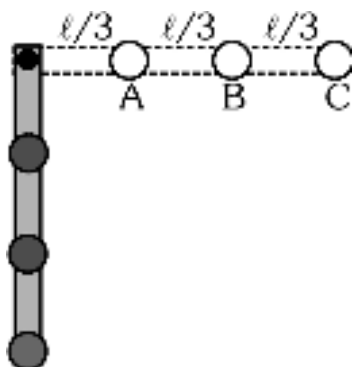
A	B	E
0	0	0
0	1	1
1	0	1
1	1	1

(D)

A	B	E
0	0	1
0	1	1
1	0	1
1	1	0

SECTION-II

1) A light rod of length (ℓ) carries three equal masses A, B and C as shown in figure. Velocity of B in vertical position of rod, if it is released from horizontal position as shown in figure is $\sqrt{\frac{ng\ell}{7}}$. Then



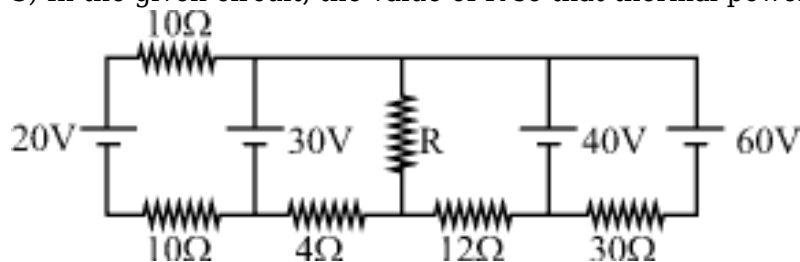
find value of n.

2) A block of metal weighing 2 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s and at a speed of 5 m/s. The initial acceleration of the block is $\frac{x}{2} \text{ m/s}^2$. The

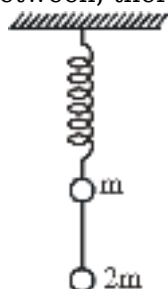
value of x is.



3) In the given circuit, the value of R so that thermal power generated in R will be maximum (in Ω).



4) A ball of mass $m = 1 \text{ kg}$ is hung from a spring of constant 400 N/m and another ball of mass $2m$ is hanging from it as shown with the help of a thread. The system is in equilibrium. Now the thread is burned. As a result, ball of mass m starts moving upwards. Eventually it will stop under the influence of gravity and spring. In between, there is a point at which its speed is maximum. What is that



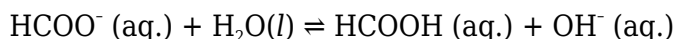
maximum speed (in m/s) ?

5) Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles $0, \pi/3, 2\pi/3$ and π . When they are superposed, the intensity of the resulting wave is nI_0 . The value of n is

PART 2 : CHEMISTRY

SECTION-I

1) The freezing point of a 0.1M formic acid aqueous solution is -0.2046°C . Find equilibrium constant of the reaction -



Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ K}\cdot\text{kg mole}^{-1}$

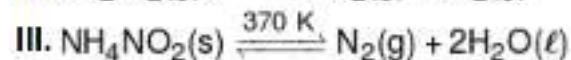
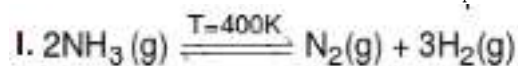
Assume solution to be very dilute

- (A) $1.1 \times 10^{-3} \text{ M}$
- (B) $9 \times 10^{-12} \text{ M}$
- (C) $9 \times 10^{-13} \text{ M}$
- (D) $1.1 \times 10^{-11} \text{ M}$

2) In which of the following set of orbitals, number of spherical nodes are same.

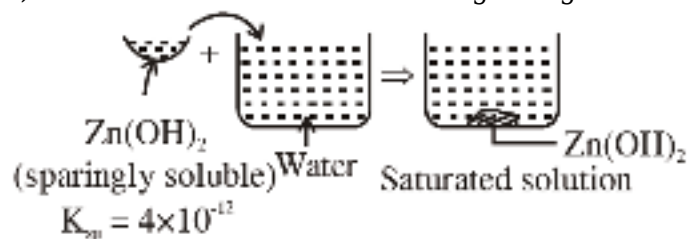
- (A) $4d_{z^2}$ & $4f$
- (B) $3d_{x^2-y^2}$ & $3p_x$
- (C) $4d_{xy}$ & $3d_{z^2}$
- (D) $3p_x$ & $4d_{x^2-y^2}$

3) Select the reactions in which $K_p > K_c$.



- (A) I, II, III
- (B) I, III
- (C) II, III
- (D) I & II

4) Select the correct statements regarding following process.



I. Solubility of Zn(OH)_2 is 10^{-4} mol/lit.

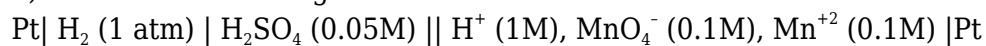
II. On addition of small quantity of OH^{-} solubility decreases

III. On addition of conc. HCl solubility increases

IV. In presence of 0.01M ZnCl_2 , solubility becomes 10^{-5} mol/lit.

- (A) I & II
- (B) I, II & IV
- (C) I & IV
- (D) I, II, III & IV

5) EMF of the following cell at 25°C is



(Given that $E_{\text{H}^+, \text{Mn}^{+2}, \text{MnO}_4^-}^{\circ} = -1.51\text{V}$ at 25°C. ($2.303RT/F = 0.06$))

- (A) 1.39 V
- (B) 1.57 V
- (C) 1.52 V
- (D) -1.42 V

6) The reaction $[\text{Fe}(\text{CN})_6]^{3-} \rightarrow [\text{FeF}_6]^{3-}$ takes place with :

- (A) Decrease in magnetic moment
- (B) Increase in magnetic moment
- (C) Decrease in co-ordination number
- (D) Increase in co-ordination number

7) Which of the following is not hydrolysed ?

- (A) CCl_4
- (B) SiCl_4
- (C) SnCl_4
- (D) PbCl_4

8) Which of the following is not planar?

- (A) H_2O
- (B) BrF_3
- (C) XeF_4
- (D) None of these

9) An excited state of H-atom emits a photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by :

- (A) $\sqrt{\lambda R(\lambda R - 1)}$
- (B) $\sqrt{\frac{\lambda R}{(\lambda R - 1)}}$
- (C) $\sqrt{\lambda R(\lambda R - 1)}$
- (D) $\sqrt{\frac{(\lambda R - 1)}{\lambda R}}$

10) Co-ordinated as well as hydrogen bonded water are present in :

- (A) $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$
- (B) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

(C) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

(D) All of these

11) How many grams of solid NaOH must be added to 100 ml of a buffer solution which is 0.1 M each w.r.t. Acid HA and salt $\text{Na}^+ \text{A}^-$ to make the pH of solution 5.5.

Given : $\text{p}K_a (\text{HA}) = 5$.

(Use $\text{antilog}(0.5) = 3.16$)

(A) 2.08×10^{-1}

(B) 3.05×10^{-3}

(C) 4.01×10^{-2}

(D) 5.19×10^{-2}

12) The correct IUPAC name of complex $[\text{Hg}(\text{CHCl}_2)(\text{Ph})]$ is :

(A) Dichloromethylbenzylmercury(II)

(B) Benzyl dichloromethylmercury(II)

(C) Chloroformphenylmercury(II)

(D) Dichloromethylphenylmercury(II)

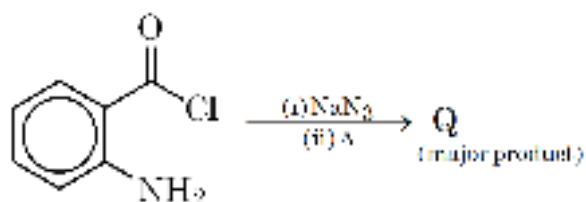
13) Which of the following species has maximum tendency to gain an electron ?

(A) H^+

(B) He^+

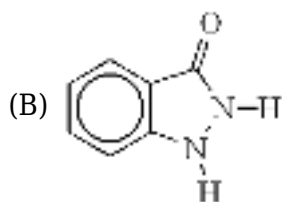
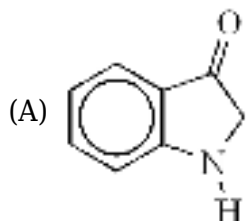
(C) Na^+

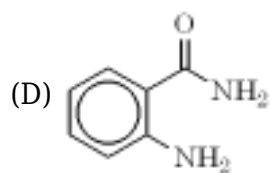
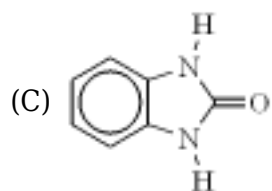
(D) Cl^+



14)

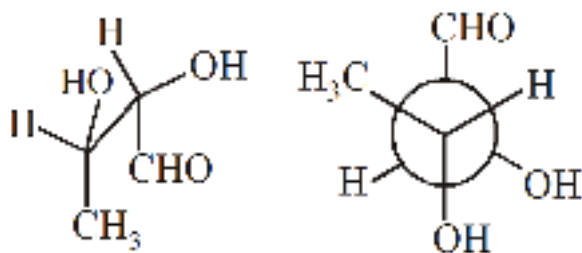
Product 'Q' is :



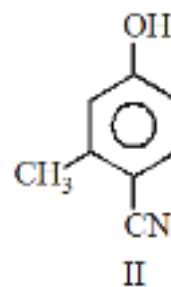
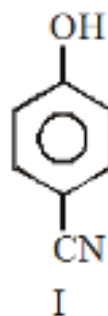


15)

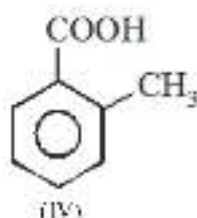
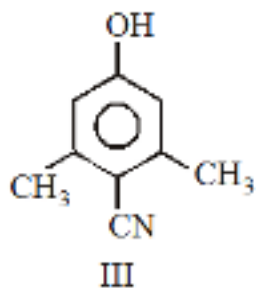
What is the relation between given compounds.



- (A) Enantiomers
- (B) Diastereomers
- (C) Conformers
- (D) Identical

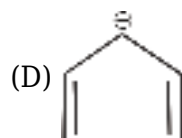
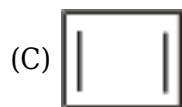
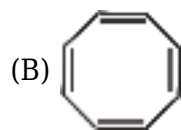
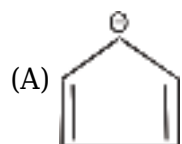


16) Which of the following correct order of acidic strength.



- (A) I > III > II > IV
- (B) II > III > I > IV
- (C) I > II > III > IV
- (D) IV > I > II > III

17) Which one is non-aromatic?



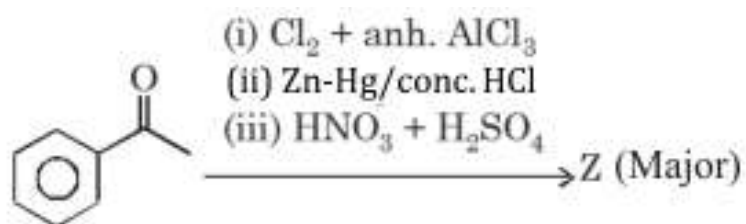
18) Which of the following nitrogen containing compound does not give Lassaigne's test ?

(A) Phenyl hydrazine

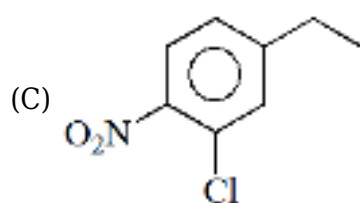
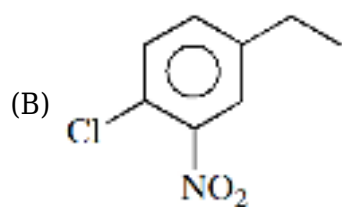
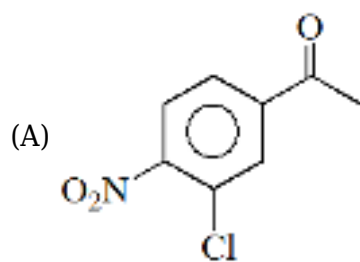
(B) Glycine

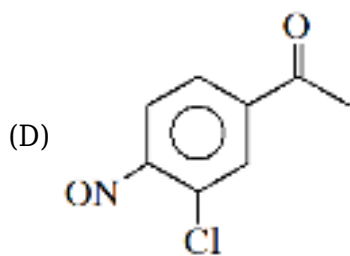
(C) Urea

(D) Hydrazine

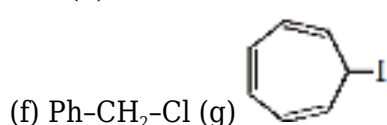
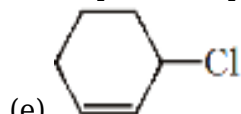
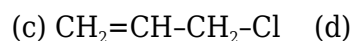
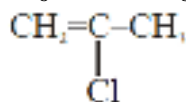
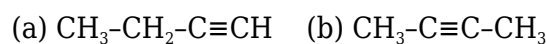


19)
Z is :





20) How many of the following will give white ppt of silver salt when treated with ammonical AgNO_3 solution?



(A) 3

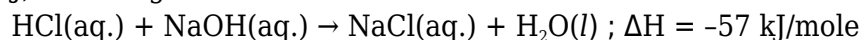
(B) 4

(C) 5

(D) 6

SECTION-II

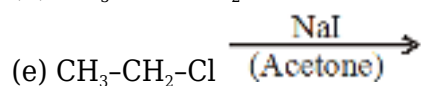
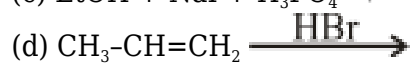
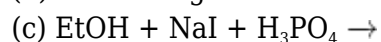
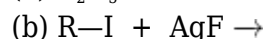
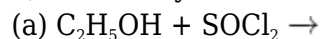
1) 200 ml of 0.1M NaOH solution is mixed with 200 ml of 0.025M H_2SO_4 solution. Calculate $|\Delta H|$ (in J) of mixing if

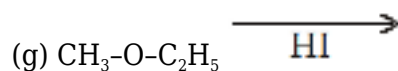
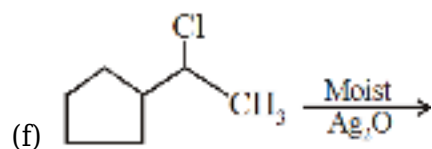


2) Two mole of an ideal gas is isothermally compressed in two stages starting from (1 atm, 20 lit) to 10 lit. at external pressure P_1 then upto 2 lit at external pressure P_2 . Find work done (lit-atm) in the process.

3) The brown ring complex compound is formulated as $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$. Find the oxidation state of Fe in the complex.

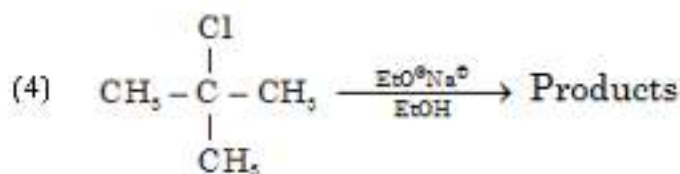
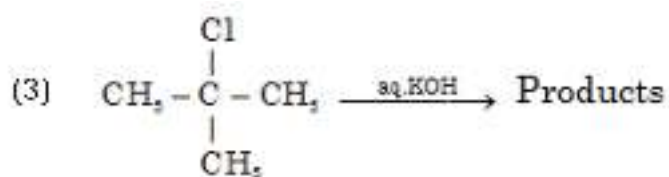
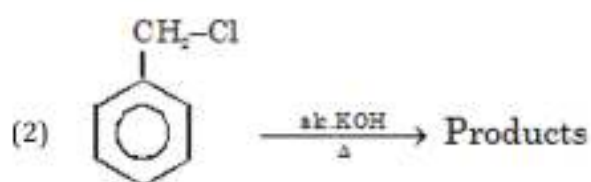
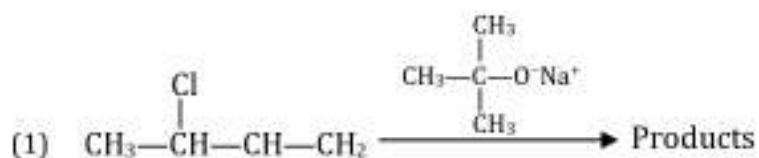
4) How many reactions giving alkyl halide as one of the final product ?





5)

How many of the following reaction(s) will give major product via elimination mechanism.()



PART 3 : MATHEMATICS

SECTION-I

1) If (x, y, z) be an arbitrary point lying on a plane P which passes through the point $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$

(A) 0

(B) 3

(C) 39

(D) -45

2) A rectangle ABCD of dimension 'a' and '2a' is folded along diagonal BD such that planes ABD and CBD are perpendicular to each other. If C point goes to C', then the distance AC' is

(A) $\sqrt{3}a$

(B) $\sqrt{85}a$

(C) $\frac{\sqrt{85}}{5}a$

(D) $\sqrt{\frac{17}{5}}a$

3) A plane P intersects 4 lines L_1, L_2, L_3 and L_4 (given below) at A,B,C,D

$$L_1 : \frac{x-3}{2} = \frac{y-3}{1} = \frac{z-3}{2}, L_2 : \frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{2},$$

$$L_3 : \frac{x}{2} = \frac{y-3}{1} = \frac{z}{2}, L_4 : \frac{x}{2} = \frac{y-3}{1} = \frac{z-3}{2},$$

then the minimum area of quadrilateral ABCD is

(A) 9

(B) $3\sqrt{2}$

(C) 3

(D) $9\sqrt{2}$

$$4) \frac{\sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}}{\sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}} \text{ equals}$$

(A) $\sqrt{2}$

(B) $\sqrt{2} - 1$

(C) $\sqrt{2} + 1$

(D) None of these

5) Let $a_1=b_1=1$ and $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}, \forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to

(A) 461

(B) 463

(C) 641

(D) 643

6) Let P_1 be a regular polygon of n sides with side length x and let P_{k+1} be the regular polygon of n sides formed by joining the mid points of consecutive sides of P_k . Also perimeters of the P_i 's form

an infinite geometric progression. If $n = 6$ and $x = \frac{1}{2 + \sqrt{3}}$, then the sum of this infinite geometric progression is :

- (A) 8
- (B) 17
- (C) 14
- (D) 12

7) For $x \geq 3$, Let $f(x) = \frac{-x + x\sqrt{4x-3}}{2}$

Suppose $a_1, a_2, a_3, \dots$ be sequence satisfying $a_1 > 3$ and $a_{2024} = 2024$ and for $n = 1, 2, 3, \dots, 2023$ Define $a_{n+1} = f(a_n)$. Then

$a_1 + \left(\sum_{i=1}^{2023} \frac{(a_{i+1})^3}{(a_{i+1})^2 + a_i a_{i+1} + (a_i)^2} \right)$ is

- (A) 2024
- (B) 4047
- (C) 2025
- (D) 4049

8) A ray of light is sent along the line $x - 2y + 5 = 0$. Upon reaching the line $3x + 2y + 7 = 0$, the ray is reflected from it. Then the equation of the line containing the reflected ray is

- (A) $19x - 22y + 79 = 0$
- (B) $22x + 19y + 11 = 0$
- (C) $11x - 13y + 9 = 0$
- (D) $19x + 22y + 11 = 0$

9) The ends A, B of a fixed straight line of length 'a' and ends A' and B' of another fixed straight line of length 'b' slide upon the axis of X & the axis of Y (one end of each line on axis of X & the other end of each line on axis of Y). Find the locus of the centre of the circle passing through A, B, A' and B'.

- (A) $(2ax + 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$
- (B) $(2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 + b^2)^2$
- (C) $(2ax - 2by)^2 + (2bx + 2ay)^2 = (a^2 - b^2)^2$
- (D) $(2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$

10) If $a + b + c = 0$, then the value of expression $a^6 + b^6 + c^6 - \frac{(a^2 + b^2 + c^2)^3}{4} - \frac{(a^3 + b^3 + c^3)^2}{3}$ is -

- (A) negative for some values of a, b & c
- (B) positive for some negative values of a, b & c
- (C) independent of values of a, b & c
- (D) independent of values of a, b & c only if $a, b, c \in \mathbb{R}^+$

11)

Consider a parabola $y = \frac{x^2}{4}$ and the point $F(0, 1)$.

Let $A_1(x_1, y_1), A_2(x_2, y_2), A_3(x_3, y_3), \dots, A_n(x_n, y_n)$ are 'n' points on the parabola such $x_k > 0$ and

$\angle OFA_k = \frac{k\pi}{2n}$ ($k = 1, 2, 3, \dots, n$ and 'O' is origin). Then the value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n FA_k$, is

- (A) $\frac{2}{\pi}$
- (B) $\frac{4}{\pi}$
- (C) $\frac{\pi}{2}$
- (D) $\frac{\pi}{4}$

12) Let the focal chord of the parabola $P: y^2 = 4x$ along the line $L: y = mx + c$, $m > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola $H: x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

- (A) $2\sqrt{6}$
- (B) $2\sqrt{14}$
- (C) $4\sqrt{6}$
- (D) $4\sqrt{14}$

13) Two curves $x^2 + y^2 + 6x - 24y + 72 = 0$ and $x^2 - y^2 + 6x + 16y - 46 = 0$ intersect at four points P, Q, R, S lying on a parabola.

Let A be the focus of the parabola, then the value of $AP + AQ + AR + AS$ is

- (A) 40
- (B) 20
- (C) 30
- (D) 50

14) If $f(x) = \tan^{-1}x - \frac{2}{\pi}(\tan^{-1}x)^2 + \frac{4}{\pi^2}(\tan^{-1}x)^3 - \dots \infty$, then the sum of integral values of a for which equation

$f^2(x) + (\sin^{-1}x)^2 = a$, possess solution(s), is

- (A) 6
- (B) 7
- (C) 9
- (D) 10

15) $\int \frac{x^2(1 - \ln x)}{(\ln x)^4 - x^4} dx$ equals

- (A) $\frac{1}{4} \ln \left(\frac{\ln x - x}{\ln x + x} \right) - \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$
 (B) $\frac{1}{2} \ln \left(\frac{x}{\ln x} \right) - \frac{1}{4} \ln (\ln^2 x - x^2) + C$
 (C) $\frac{1}{4} \ln \left(\frac{\ln x + x}{\ln x - x} \right) + \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$
 (D) $\frac{1}{4} \left(\ln \left(\frac{\ln x - x}{\ln x + x} \right) + \tan^{-1} \left(\frac{\ln x}{x} \right) \right) + C$

Where C is an arbitrary constant.

16) If z_1, z_2 are complex numbers such that

$\operatorname{Re}(z_1) = |z_1 - 1|, \operatorname{Re}(z_2) = |z_2 - 1|$ and

$\arg(z_1 - z_2) = \frac{\pi}{6}$, then $\operatorname{Im}(z_1 + z_2)$ is equal to:

- (A) $\frac{\sqrt{3}}{2}$
 (B) $\frac{2}{\sqrt{3}}$
 (C) $\frac{1}{\sqrt{3}}$
 (D) $2\sqrt{3}$

17) Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set

$X = \left\{ z \in \mathbb{C} : \operatorname{Re}(az^2 + bz) = a \text{ and } \operatorname{Re}(bz^2 + az) = b \right\}$ is equal to

- (A) 1
 (B) 3
 (C) infinite
 (D) 2

18) Maximum and minimum value of $\frac{6|z|^2 - z^2 - (\bar{z})^2 - 8}{8}$ are m_1 and m_2 respectively, if z moves on $z\bar{z} - \sqrt{2}\bar{z}i + \sqrt{2}iz = 0$, then $(m_1 + m_2)$ equal

- (A) 4
 (B) 5
 (C) 6
 (D) 7

19) Let the set of complex numbers $(a_1, b_1); (a_2, b_2), (a_3, b_3) \dots$ denoting the points on the complex plane satisfying $(a_{n+1}, b_{n+1}) = (\sqrt{3}a_n - b_n, \sqrt{3}b_n + a_n)$ for $n = 1, 2, 3, \dots$. Suppose

$(a_{100}, b_{100}) = (2, 4)$ then the value of $(a_1 + b_1)$ is equal to

- (A) $\frac{1}{2^{96}}$
 (B) $\frac{1}{2^{97}}$
 (C) $\frac{1}{2^{98}}$
 (D) $\frac{1}{2^{99}}$

$$f(x) = \begin{cases} (\sin x) [\sin x]^2 \log_{(1+\sin x)} 2 & -\frac{\pi}{2} < x < 0 \\ \frac{\ln(e^{\tan^2 x + 2\sqrt{\{\tan x\}}})}{\tan(\sqrt{\tan x})} & 0 < x < \frac{\pi}{2} \end{cases}$$

20) Consider (where $[.]$ and $\{.\}$ are greatest integer and fractional part function respectively), then-

- (A) $f(0) = \pi^2 \Rightarrow f(x)$ is continuous at $x = 0$
 (B) $f(0) = 2 \Rightarrow f(x)$ is continuous at $x = 0$
 (C) $f(0) = e^2 \Rightarrow f(x)$ is continuous at $x = 0$
 (D) $f(x)$ has irremovable discontinuity at $x = 0$

SECTION-II

1) Find the number of ordered pair(s) (a, b) for which the function $f(x) = \text{sgn}((x^2 - ax + 1)(bx^2 - 2bx + 1))$ is non differentiable at exactly one point (where a, b are integer).
 [Note : $\text{sgn}(x)$ denotes signum function of x .]

2) If $f(x) = \begin{cases} \frac{1-\sin x}{(\pi-2x)^2} \cdot \frac{\ln \sin x}{\ln(1+\pi^2-4\pi x+4x^2)} & , x \neq \frac{\pi}{2} \\ k & , x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \pi/2$, then $\left| \frac{1}{k} \right|$ is

3) Let $\frac{\pi}{3} < A < \frac{\pi}{2}$ and a function $f(x)$ is defined as:

$$\begin{cases} \lim_{n \rightarrow \infty} \frac{ax^2(\sin A - \sin^3 A) - (5x - b)|\sin A - \sin^3 A|^n}{(\sin A - \sin^3 A) - |\sin A - \sin^3 A|^n}, x \in \mathbb{Q} \\ \lim_{n \rightarrow \infty} \frac{ax^2(\sin A + \sin^3 A) + (5x - b)|\sin A + \sin^3 A|^n}{(\sin A + \sin^3 A) + |\sin A + \sin^3 A|^n}, x \notin \mathbb{Q} \end{cases}$$

If $f(x)$ is continuous at $x = 2$ and $x = 3$ then the value of $b - a$, is

4) Let 'T' be a tangent line at the point $T\left(t, \frac{t^2}{2}\right)$ ($t > 0$) on the curve $C: y = \frac{x^2}{2}$. A circle S touches the curve C at T and the x-axis. Denote by $(x(t), y(t))$ the centre of circle S. If

$$\lim_{r \rightarrow 0^+} \int_r^1 x(t)y(t)dt = \frac{\tan \frac{3\pi}{8}}{6a}, \text{ then the value of } a \text{ is}$$

5) Let $f'(x) = -\frac{f(x)}{\sqrt{f^2(x) + g^2(x)}}$ & $g'(x) = 1 - \frac{g(x)}{\sqrt{f^2(x) + g^2(x)}}$ $\forall x \in \mathbb{R}$, $g(0) = 0$, $f(0) = 10$,
 $f(x), g(x) > 0 \forall x \in \mathbb{R}$ and $\lim_{x \rightarrow \infty} f(x) = 0$, then the value of $\lim_{x \rightarrow \infty} g(x)$ is

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	C	A	A	C	D	A	A	A	A	A	C	C	B	B	A	C	B	A	A	B

SECTION-II

Q.	21	22	23	24	25
A.	8	5	3	1	3

PART 2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	B	D	A	D	B	B	A	D	B	C	A	D	B	C	B	D	B	D	C	B

SECTION-II

Q.	46	47	48	49	50
A.	570	100	1	6	2

PART 3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	B	C	C	C	A	D	B	A	D	C	B	B	A	D	A	D	C	C	C	D

SECTION-II

Q.	71	72	73	74	75
A.	6	64	5	5	5

SOLUTIONS

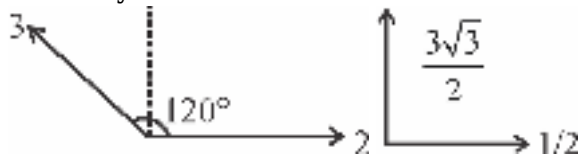
PART 1 : PHYSICS

1) **Explain Question :** The question asks about the velocity of a swimmer with respect to ground or w.r.t. river.

Concept : $\vec{V}_{mr} = \vec{V}_m - \vec{V}_f$

Solution :

WRT Boy



$$V_{mb} = \frac{\sqrt{27+1}}{2} = \sqrt{7} < 3$$

WRT Girl



$$V_{mg} = 3 \text{ km/hr.}$$

Final Answer : Option (C) is correct

2)

Question Explanation:- The question asks about the position of a particle under one-dimensional motion when its velocity is 0.

Concept:- This question is based on one dimension motion with variable acceleration.

Solution :-

$$x = 4a \left(t + a \sin \left(\frac{t}{a} \right) \right)$$

$$\frac{dx}{dt} = \frac{d(4at)}{dt} + \frac{d}{dt} [4a^2 \sin(t/a)]$$

$$0 = 4a + 4a^2 \frac{1}{a} \left(\cos \left(\frac{t}{a} \right) \right)$$

$$\frac{t}{a} = \pi$$

$$t = a\pi$$

$$x = 4a^2\pi \text{ metre}$$

Final Answer: Option (A) is correct.

3) **Explain Question :** We need to find the number of drops falling of a ball during projectile motion.

$$T = \sqrt{\frac{2h}{g}}$$

Concept :

Solution :

Time of flight of wet

$$= \sqrt{\frac{2 \times 31.25}{9.8}} = 2.6 \text{ TOF will not change by reducing mass. Hence only two drop will hit}$$

Final Answer : Option (B) is correct

4) **Explain Question :** The question needs to find the initial velocity of conveyer belt when the block closes to slip on it.

Concept : When slipping stops $V_{\text{net}} = 0 \Rightarrow \vec{V}_{\text{net}} = \vec{u}_{\text{net}} + \vec{a}_{\text{net}}$

Solution :

$$a = mg \Rightarrow v = u + at$$

Final Answer : Option (C) is correct

5) **Explain Question :** We need to find work done by a given force during planar motion of an object.

Concept :
$$W = \int \vec{F} \cdot d\vec{r}$$

Solution :

$$\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$$

$$x^2 = y$$

$$x = y^{1/2}$$

Substitute in above equation

$$\vec{F} = (3y^{3/2} - 5z)\hat{j} + 4z\hat{k}$$

$$dw = F \cdot dx \text{ In } (\hat{j}) \text{ direction}$$

$$W = \int_0^4 3y^{3/2} dy$$

$$\rightarrow \left[\frac{3y^{5/2}}{5/2} \right]_0^4$$

$$W = \frac{192}{5} \text{ Joule}$$

(in 'z-direction' work done is always zero)

Final Answer : Option (D) is correct

6) **Explain Question :** The questions asks to compare Young's modulus & ductility of two wires under stress.

Concept : $Y = \tan \theta$ of stress strain curve.

Solution :

$$Y = \frac{\text{stress}}{\text{strain}} \text{ for same strain } (\text{stress})_A > (\text{stress})_B$$

$$Y_A > Y_B$$

Breaking point of B > Breaking point of A

So, A is less ductile.

Final Answer : Option (A) is correct

7) **Explain Question :** Question asks to find diameter of wire using vernier callipers.

Concept : True reading = Measured value + zero error.

Solution :

From screw gauge z.e. = + 0.20 mm

Dia of wire = 3.5 - 0.2 = 3.3 mm = 0.33 cm

For vernier callipers, z.e. = - 0.05 cm

Thus it must show 0.33 - 0.05 = 0.28 cm

so (a) is correct

Final Answer : Option (A) is correct

8) **Explain Question :** The question asks to find Young's modulus of a wire under searle's setup.

Concept : $Y = \frac{F\ell}{A\Delta\ell}$

Solution :

$$Y = \frac{F\ell}{A\Delta\ell}$$

$$\Delta\ell = \frac{1}{4}\text{mm} = 0.25 \times 10^{-3} \text{ m}$$

$$F = 50 \times 10 \text{ N} = 500 \text{ N.}$$

$$l = 3 \text{ m}$$

$$A = 10^{-5} \text{ m}^2$$

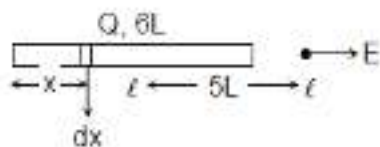
$$\therefore Y = \frac{5 \times 10^2 \times 3}{10^{-5} \times 0.25 \times 10^{-3}} = 6 \times 10^{11} \text{ N/m}^2$$

Final Answer : Option (A) is correct

9) **Explain Question :** We need to find net electric field at a point due to two thin uniformly charged wires.

Concept : $\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2$

Solution :



E due rod.

$$= \int_0^{6L} \frac{1}{4\pi\epsilon_0} \cdot \left(\frac{Q}{6L}\right) \frac{dx}{(8L-x)^2} = \frac{1}{64\pi\epsilon_0} \frac{Q}{L^2}$$

$$E_{\text{net}} = 2 E \cos 30^\circ.$$

Final Answer : Option (A) is correct

10) **Explain Question :** The questions asks about the magnetic field component of a em wave.

Concept : $\hat{V} = \hat{E} \times \hat{B}$

Solution :

Wave is propagating in -ve x-direction. $\therefore \hat{E} \times \hat{B} = \hat{V}$

$$\hat{K} \times \hat{B} = -\hat{i}$$

$$\text{So } \hat{B} = \hat{j}$$

$$B_0 = \frac{E_0}{C} = \frac{1.5 \times 10^6}{3 \times 10^8} = 5 \times 10^{-3} \text{ T}$$

$$B = (B_0 \hat{j}) \sin[\omega t + kx + \phi]$$

Or

$$\vec{B} = (B_0 \hat{j}) \cos[\omega t + kx + \phi]$$

$$\vec{B} = (5 \times 10^{-3} \hat{j}) \cos(\omega t + kx)$$

Final Answer : Option (A) is correct

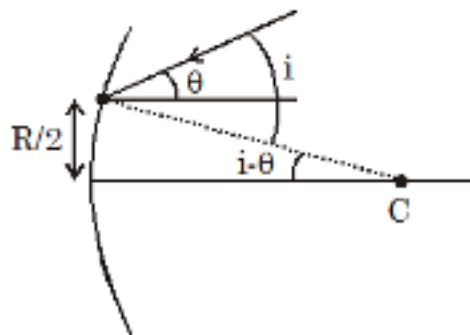
11) **Explain Question :** The question asks about the angle of incidence for TIR.

Concept : $\sin c = \frac{1}{\mu}$

Solution :

The ray will be totally reflected when angle of incidence is equal to critical angle.

$$\frac{\sin i}{\sin \frac{\pi}{2}} = \frac{\mu_2}{\mu_1} \Rightarrow \sin i = \frac{\mu_2}{\mu_1} = \frac{1}{2} \Rightarrow i = 30^\circ$$



From figure, $i - \theta = 30^\circ \Rightarrow \theta = 0$

Final Answer : Option (C) is correct

12)

Explain Question : It is asked to predict the nature of image using $\frac{1}{v}$ vrs $\frac{1}{u}$ graph of convex mirror.

Concept : $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Solution :

$f = \oplus$ (Convex mirror)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

at point A

$$u > 0 \quad \& \quad v > 0$$

$$\& \quad \frac{1}{v} > \frac{1}{u} \Rightarrow u > v$$

Magnification $m = -\frac{v}{u}$ is Negative & Less than one

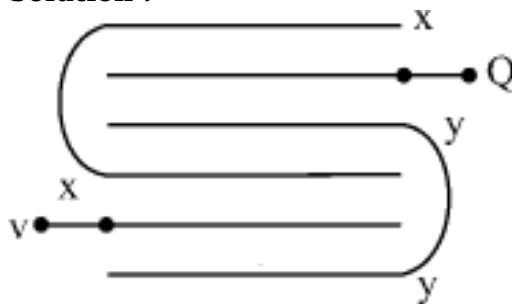
So, Image is virtual, inverted, diminished.

Final Answer : Option (c) is correct

13) **Explain Question :** The question asks about the c_{eq} in case of multiple parallel plane arrangement.

Concept : $C = \frac{A\epsilon_0}{d}$

Solution :



]

$$C(y - V) + C(y - x) + Cy = 0$$

$$3y - x = V$$

$$C(x - 0) + C(x - y) + C(x - V) = 0$$

$$3x - y = V$$

$$\Rightarrow 8x = 4V \Rightarrow x = \frac{V}{2}, y = \frac{V}{2}$$

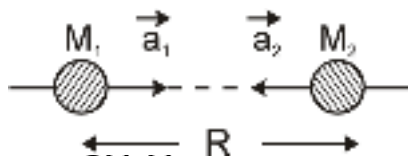
$$\Rightarrow \Delta Q = \frac{\epsilon_0 A}{d} V$$

Final Answer : Option (B) is correct

14) **Explain Question :** You need to find relative acceleration of one object wrt another under mutual gravitational attraction.

Concept : $a = \frac{F}{m} = \frac{Gm}{r^2}$ $\vec{a}_{\text{net}} = \vec{a}_1 - \vec{a}_2$

Solution :



$$\vec{a}_1 = \frac{GM_1 M_2 / M_1}{R^2}$$

$$\vec{a}_2 = -\frac{GM_1 M_2 / M_2}{R^2}$$

acceleration of M_1 w.r.t. M_2

$$\vec{a}_{\text{rel.}} = \vec{a}_1 - \vec{a}_2 = \frac{G(M_1 + M_2)}{R^2} = \frac{GM}{R^2}$$

Final Answer : Option (B) is correct

15) **Explain Question :** It is asked to find ratio of power radiated by two spherical bodies at

Concept : $P = eA\sigma T^4$; $\lambda_m T = b$

Solution :

$$P = eA\sigma T^4$$

$$\frac{P_1}{P_2} = \frac{1}{1} \times \frac{6^2}{18^2} \times \left(\frac{T_1}{T_2}\right)^4$$

$$\frac{P_1}{P_2} = \frac{1}{9} \times 3^4 = 9$$

$$T_1 \lambda_1 = T_2 \lambda_2$$

$$\frac{T_1}{T_2} = \frac{1500}{500} = 3$$

Final Answer : Option (A) is correct

16) **Explain Question :** It is asked to compare internal energies of gases and speed of sound in them.

Concept : $U = \frac{f}{2}RT$; $V = \sqrt{\frac{\gamma RT}{M}}$ **Solution :**

$$U = \frac{1}{2}f n r T = \frac{f r T}{2}$$

□ A and B are wrong.

$$V_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\left(\frac{2}{f} + 1\right) \frac{RT}{M}}$$

⇒ more 'f', less 'v'

$$\therefore V_5 > V_7$$

Final Answer : Option (C) is correct

17) **Explain Question :** It is asked to find the ratio of distance of two bar magnet from a given point where a compass needles placed.

Concept : $B_{\text{eq}} = \frac{KM}{r^3}$; $E_{\text{Axial}} = \frac{2kM}{r^3}$

Solution :

According to tangent law at point 'P'

$$B_{\text{axis}} = B_{\text{eq}} \tan \theta \Rightarrow \frac{2kM}{r_2^3} = \frac{kM}{r_1^3} \tan \theta$$

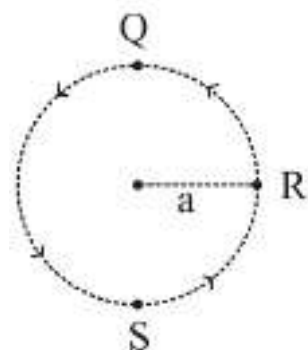
$$\Rightarrow \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{\tan \theta}{2}\right) \Rightarrow \frac{r_1}{r_2} = \left(\frac{\tan \theta}{2}\right)^{1/3}$$

Final Answer : Option (B) is correct

18) **Explain Question :** Question asks about $\int \vec{B} d\vec{\ell}$ over a certain portion of amperian loop.

Concept : $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i_{\text{en}}$

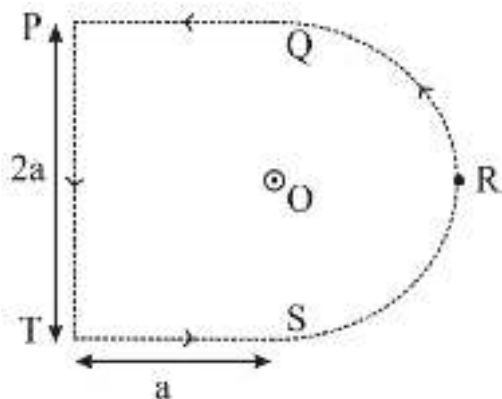
Solution :



For the circular Amperian loop

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i \Rightarrow \int_{SRQ} \vec{B} \cdot d\vec{\ell} = \mu_0 i$$

$$\therefore \int_{SRQ} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{2}$$



For given amperian loop

$$\oint \vec{B} \cdot d\vec{\ell} = \int_{SRQ} \vec{B} \cdot d\vec{\ell} + \int_{QPTS} \vec{B} \cdot d\vec{\ell}$$

$$\mu_0 i = \frac{\mu_0 i}{2} + \int_{QPTS} \vec{B} \cdot d\vec{\ell}$$

$$\int_{QPTS} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{2}$$

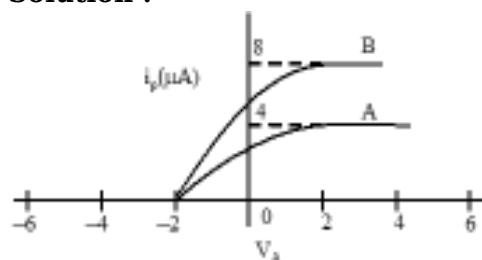
QPTS

Final Answer : Option (A) is correct

19) **Explain Question :** The question asks to draw photocurrent vrs anode voltage graph of a photoelectric effect.

Concept : $eV = hf - \Phi_0$

Solution :



Stopping potential, $V = (hn - W)/e = 2V$

In each case, the stopping potential is 2 V as energy of incident photon is the same. As the photon intensity is doubled, the saturation current is also doubled. The variation of i_p against V_a is sketched in the figure

Final Answer : Option (A) is correct

20) **Explain Question :** Question needs to find truth table for certain logic gate arrangement.

Concept : Use of Boolean algebra.

Solution :

Theoretical

Final Answer : Option (B) is correct

21) **Explain Question :** The question asks about velocity of a point of a rigid body under pure rotation

Concept : Energy conservation $\frac{1}{2}I\omega^2 = \Delta U = mgh$

Solution :

Loss in P.E. = Gain in K.E.

$$\begin{aligned}
 & mg\frac{\ell}{3} + mg\left(\frac{2\ell}{3}\right) + mg\ell \\
 &= \frac{1}{2} \left(m\left(\frac{\ell}{3}\right)^2 + m\left(\frac{2\ell}{3}\right)^2 + m\ell^2 \right) \omega^2 \\
 \Rightarrow \omega &= \sqrt{\frac{36g}{14\ell}} \Rightarrow v_B = \omega \ell_B = \frac{2\ell}{3} \sqrt{\frac{36g}{14\ell}} = \sqrt{\frac{8g\ell}{7}} = 8
 \end{aligned}$$

Final Answer : (8)

22) **Explain Question :** It is asking to find acceleration of a block under the force of a water jet.

Concept :
$$a = \frac{f}{m} \left(\frac{v}{m} \frac{dm}{dt} \right)$$

Solution :

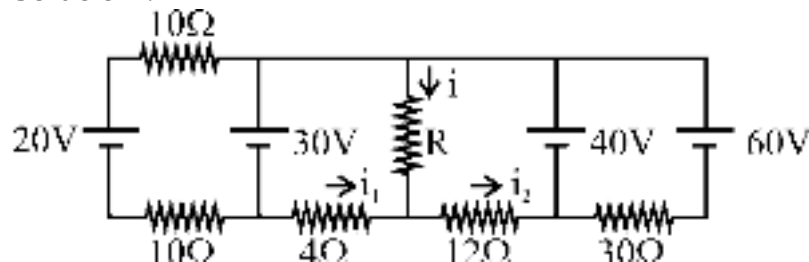
$$2a = v \frac{dm}{dt} = 5 \times 1$$

Final Answer : (5)

23) **Explain Question :** The questions asks about the value of resistance R in a circuit when it consumes maximum power.

Concept : Use of maximum power transfer theorem.

Solution :



$$\begin{aligned}
 R_{eq} &= R_{in} \\
 &= \frac{4 \times 12}{4 + 12} = 3\Omega
 \end{aligned}$$

Final Answer : (3)

24) **Explain Question :** Question asks about V_{max} of an object under SHM

Concept : $V_{max} = \omega A$

Solution :

$$3mg = kx_i, mg = kx_0$$

maximum speed is at equilibrium

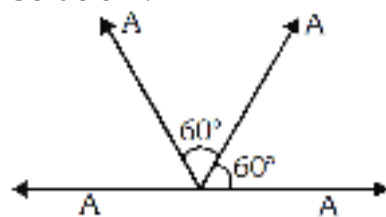
$$\begin{aligned}
 &= A\omega = \frac{2mg}{k} \times \sqrt{\frac{k}{m}} = 2g\sqrt{\frac{m}{k}} \\
 &= 20 \times \sqrt{\frac{1}{400}} = 1\text{m/s}
 \end{aligned}$$

Final Answer : (1)

25) **Explain Question :** The question asks to find resultant intensity of four waves interfering from coherent sources.

Concept : Use superposition principle.

Solution :



$$A_{\text{res}} = \sqrt{3} A$$

$$I_{\text{res}} = 3I_0$$

Final Answer : (3)

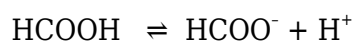
PART 2 : CHEMISTRY

26)

Explain Question: Calculate equilibrium constant of the reaction

Concept: Depression in F.P

Solution:



$$(0.1 - x) \quad \quad x \quad \quad x$$

$$\Delta T_f = i k_f m$$

$$\Delta T_f = 0.2046 = 1.86 \times (0.1 + x)$$

$$(0.1 + x) = \frac{0.2046}{1.86}$$

$$x = 0.01$$

$$K_h = \frac{K_w}{K_a}$$

$$\therefore K_a = \frac{x^2}{0.1 - x} = \frac{10^{-2}}{9}$$

$$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{\frac{10^{-2}}{9}} = 9 \times 10^{-12}$$

$\Rightarrow$

Final Answer: (B)

27) **Explain Question:** Identify which set has equal no. of spherical nodes.

Concept: No. of spherical nodes = $n - l - 1$

Solution:

$$\text{No. of spherical nodes} = n - l - 1$$

$$\Rightarrow 3 - 1 - 1 = 1 \text{ for } 3p_x$$

$$\Rightarrow 4 - 2 - 1 = 1 \text{ for } 4d_{x^2 - y^2}$$

Final Answer: (D)

28) **Explain Question:** Identify reaction for which $K_p > K_c$

Concept: $K_p = K_c(RT)^{\Delta n_g}$

Solution:

$$K_p = K_c(RT)^{\Delta n_g}$$

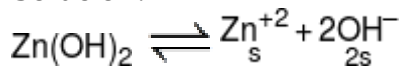
$$K_p > K_c \text{ if } \Delta n_g > 0 \text{ \& T} > \frac{1}{R} ; \left\{ \frac{1}{R} = 12.19 \right\}$$

Final Answer: (A)

29) **Explain Question:** Identify correct statement of given process.

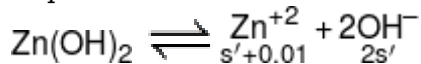
Concept: Solubility product

Solution:



$$S = (k_{\text{sp}} / 4)^{1/3} = 10^{-4}$$

In presence of 0.01 M Zn^{+2} .



$$k_{\text{sp}} = 0.01 \times (2s')^2$$

$$S' = \sqrt{(4 \times 10^{-12}) / (10^{-2} \times 4)} = 10^{-5} \text{ M}$$

on addition of OH^{-} , solubility decreases. on addition of H^{+} , Solubility increases.

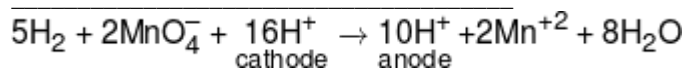
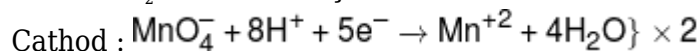
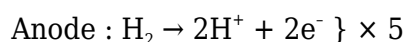
Final Answer: (D)

30) **Explain Question:** Calculate EMF of given cell

Concept: Nernst equation.

Solution:

cell reaction



$$E_{\text{cell}} = \left[E_{\text{MnO}_4^{-}/\text{Mn}^{2+}}^{\circ} - E_{\text{H}^{+}/\text{H}_2}^{\circ} \right] - \frac{0.06}{10} \frac{[\text{H}^{+}_{\text{Anode}}]^{10} [\text{Mn}^{2+}]^2}{[\text{P}_{\text{H}_2}]^5 [\text{MnO}_4^{-}]^2 [\text{H}^{+}_{\text{Cathode}}]^{16}}$$

$$E_{\text{cell}} = (1.51 - 0) - \frac{0.06}{10} \log \frac{(0.1)^{10} (0.1)^2}{(1)^{16} (0.1)^2}$$

$$E_{\text{cell}} = 1.51 - \frac{0.06}{10} \log 10^{-10}$$

$$E_{\text{cell}} = 1.57 \text{ V}$$

Final Answer: (B)

31) **Explain Question:** Identify the change observed in the complex formed.

Concept: Ligand field strength.

Solution:

CN⁻ is strong field ligand due to which Fe^{+3} configuration will be t_{2g}^5 hence 1 unpaired electron whereas in $[\text{FeF}_6]^{3-}$, F⁻ is weak field ligand hence due to weak field splitting, configuration will be $t_{2g}^3 e_g^2$ i.e. 5 unpaired electrons.

Final Answer: (B)

32) **Explain Question:** Identify compound which is not hydrolysed

Concept: Concept of Lewis acid

Solution:

The intermediate formed in reaction of tetrahalides with H_2O is sp^3d hybridised and carbon does NOT have vacant d-orbitals.

Final Answer: (A)

33) **Explain Question:** Identify non-planar molecule.

Concept: VSEPR Theory

Solution:

H_2O : triatomic and hence planar

BrF_3 : Bent T-shaped, planar

XeF_4 : Square planar

Final Answer: (D)

34)

Explanation Question: Find principal quantum number of excited state

Concept: This question is based on line spectrum of hydrogen

Solution:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right); n_1 = 1, n_2 = ?;$$

$$\frac{1}{\lambda} = R \left(\frac{1}{1} - \frac{1}{n_2^2} \right) \Rightarrow n_2^2 = \frac{R\lambda}{R\lambda - 1}$$

$$\Rightarrow n_2 = \sqrt{\frac{\lambda R}{\lambda R - 1}}$$

Final Answer

The correct option is (2)

35) **Explain Question:** Identify compound in which co-ordinated as well as hydrogen bonded water are present

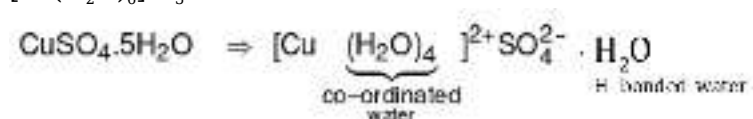
Concept: $BaCl_2 \cdot 2H_2O$: interstitial water

$[Cr(H_2O)_6]Cl_3$: co-ordinated water

Solution:

$BaCl_2 \cdot 2H_2O$: interstitial water

$[Cr(H_2O)_6]Cl_3$: co-ordinated water



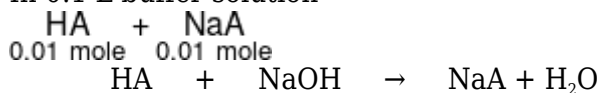
Final Answer: (C)

36) **Explain Question:** Calculate the amount of NaOH to make a solution of pH = 5.5

Concept: Buffer solution

Solution:

In 0.1 L buffer solution



$$t = 0 \quad 0.01 \quad x \text{ mole}$$

$$t \quad 0.01 - x \quad 0 \quad 0.01 + x$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$5.5 = 5 + \log \frac{0.01 + x}{0.01 - x}$$

$$0.5 = \log \frac{0.01 + x}{0.01 - x}$$

$$0.5 = \log \frac{0.01 + x}{0.01 - x}$$

$$0.0316 - 3.16x = 0.01 + x$$

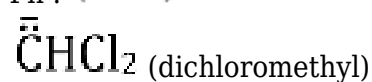
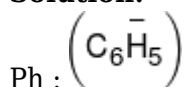
$$4.16x = 0.0216$$

$$x = \frac{2.16 \times 10^{-2}}{4.16}$$

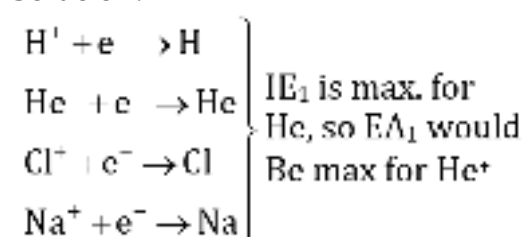
$$x = \frac{2.16}{4.16} \times 10^{-2}$$

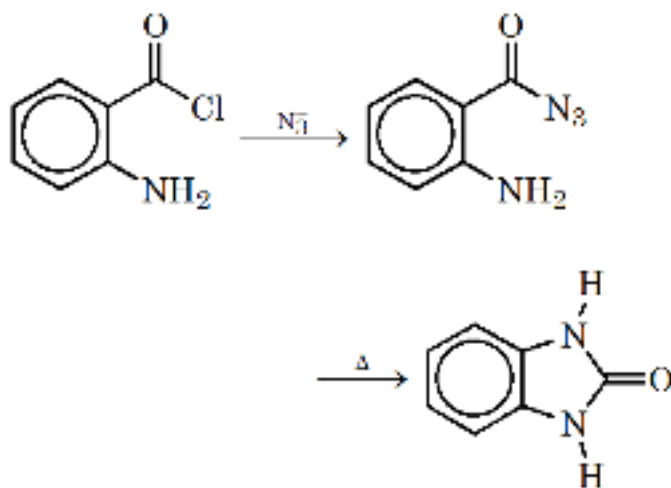
$$\frac{w}{40} = \frac{2.16}{4.16} \times 10^{-2}$$

$$w = 2.08 \times 10^{-1} \text{ g}$$

Final Answer: (A)37) **Explain Question:** Find correct IUPAC name of complex $[\text{Hg}(\text{CHCl}_2)(\text{Ph})]$ **Concept:** IUPAC naming of co-ordination compounds.**Solution:**

ox. state of central atom = +2

Final Answer: (D)38) **Explain Question:** Identify species having maximum tendency to gain an electron**Concept:** Electron Affinity**Solution:****Final Answer: (B)**39) **Explain Question:** Identify major product Q**Concept:** Substitution is done by azide**Solution:**



Final Answer: (C)

40) **Explain Question:** Relation between given compounds.

Concept: Concept of stereoisomerism and conformers.

Solution:

They are not a mirror image of each other, hence diastereomers

Final Answer: (B)

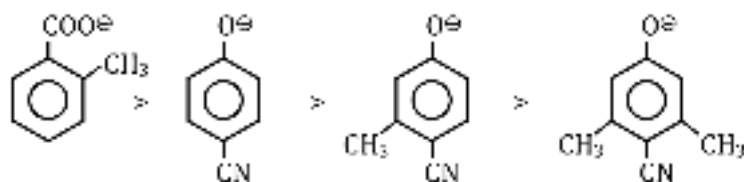
41) **Explain Question:** Identify correct order of acidic strength

Concept: Acidic strength $\propto$ stability of conjugate base

Solution:

Acidic strength $\propto$ stability of conjugate base

Stability of conjugate base is as follows



Final Answer: (D)

42) **Explain Question:** Identify non-aromatic compound

Concept: Concept of Aromaticity

Solution:

A = Aromatic

B = Non-aromatic (exist as nonplanar structure)

C,D = Antiaromatic

Final Answer: (B)

43) **Explain Question:** Identify nitrogen containing compound does not give Lassaigne's test

Concept: Concept of Lassaigne's test.

Solution:

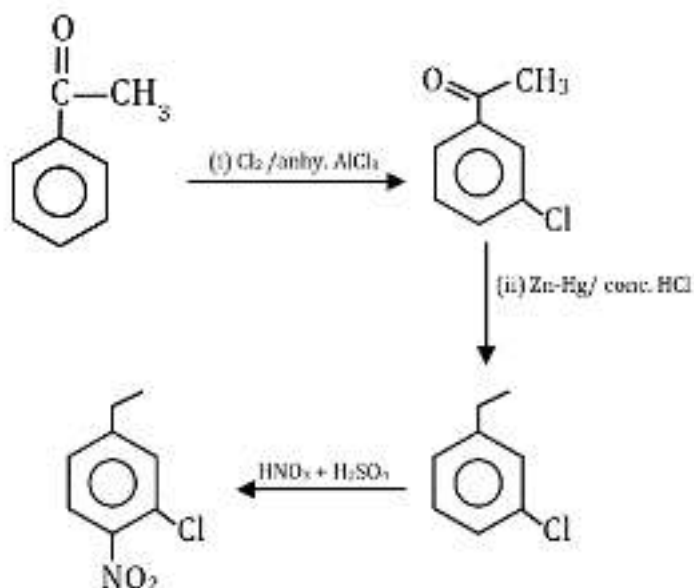
Hydrazine does not give Lassaigne's test because it has no carbon atom.

Final Answer: (D)

44) **Explain Question:** Identify major product 'Z'

Concept: ESR is followed by nitration

Solution:



Final Answer: (C)

45) **Explain Question:** Identify the compounds which will give white ppt of silver salt when treated with ammoniacal AgNO_3 solution

Concept: Tollen's Test

Solution:

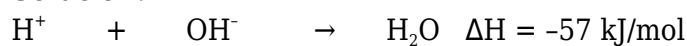
(a), (c), (e) and (f)

Final Answer: (B)

46) **Explain Question:** Calculate $|\Delta H|$ of mixing for given solutions.

Concept: Neutralization Reaction

Solution:



10 m mol 20 m mol

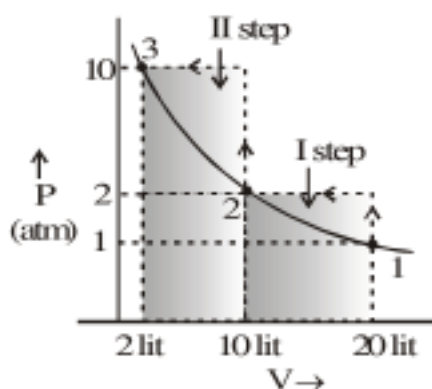
$$|\Delta H| = 10 \times 10^{-3} \times 57 \times 10^3 = 570 \text{ J}$$

Final Answer: (570)

47) **Explain Question:** Calculate work done for given process.

Concept: Work done in isothermal process.

Solution:



If vol. is 10 lit.

$$P_2 = \frac{1 \times 20}{10} = 2 \text{ atm}$$

If $V = 2 \text{ lit.}$

$$\text{then } P_3 = \frac{2 \times 10}{2} = 10 \text{ atm}$$

$$W = (20-10) \times 2 + (10-2) \times 10 \\ = 100 \text{ lit - atm}$$

Final Answer: (100)

48) **Explain Question:** Calculate oxidation state of Fe in brown ring complex

Concept: In Brown ring $\text{N}^{\oplus}\text{O}$ is present

Solution:

Iron has +1 oxidation state in brown ring complex $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$

Final Answer: (1)

49) **Explain Question:** Which reaction gives alkyl halide as one of the final product

Concept: addition of HBr, Substitution reaction

Solution:

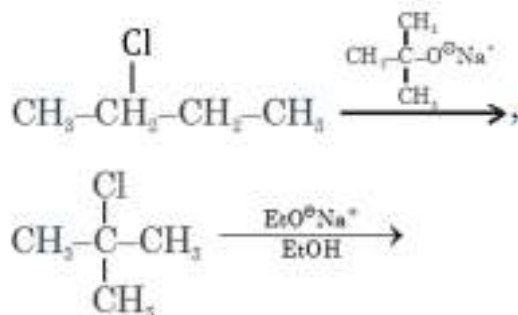
(a), (b), (c), (d), (e), (g)

Final Answer: (6)

50) **Question Explanation:** Identifying reactions that give an elimination product.

Concept: S_N vs. Elimination reactions.

Solution: Whether a reaction gives a major product via an elimination or a substitution mechanism depends on the substrate, nucleophile/base, and reaction conditions.



Final Answer: 4.

PART 3 : MATHEMATICS

51) **Explain Question :** A plane passing through three given points and a point (x, y, z) lie on plane

Concept : Intercept form of plane, symmetric expressions.

Solution :

Plane passing through (42,0,0), (0,42,0), (0,0,42)

From intercept form, equation of plane is $x + y + z = 42$

$$\Rightarrow (x-11) + (y-19) + (z-12) = 0$$

$$\text{Let } a = x - 11, b = y - 19, c = z - 12$$

$$a + b + c = 0$$

Now, given expression is

$$3 + \frac{a}{b^2c^2} + \frac{b}{a^2c^2} + \frac{c}{a^2b^2} - \frac{42}{14abc}$$

$$3 + \frac{a^3 + b^3 + c^3 - 3abc}{a^2b^2c^2}$$

If $a + b + c = 0$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow 3$$

Final Answer : (B)

52) **Explain Question :** Reflection of point across folded diagonal; coordinate geometry yields.

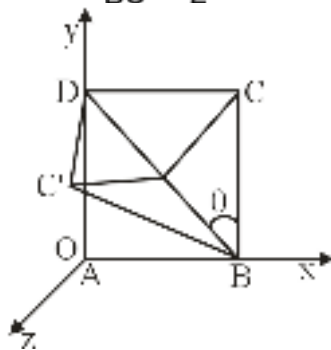
Concept: Concept : 3D reflection, foot of perpendicular.

Solution :

$$AB = CD = a$$

$$BC = AD = 2a$$

$$\tan \theta = \frac{CD}{BC} = \frac{1}{2}$$



N foot of perpendicular on BC drawn from C.

$$\sin \theta = \frac{CN}{BC} \Rightarrow \frac{1}{\sqrt{5}} = \frac{CN}{2a}$$

$$CN = C'N = \frac{2}{\sqrt{5}}a$$

equation of BD in XY plane $\frac{x}{a} + \frac{y}{2a} = 1$

and C is (a, 2a)

$$\text{coordinate of N are } \frac{x-a}{2} = \frac{y-2a}{1} = \frac{-(2a+2a-2a)}{5}$$

$$x = \frac{a}{5} \text{ and } y = \frac{8a}{5}$$

$$\text{Coordinate C' } \left(\frac{a}{5}, \frac{8a}{5}, \frac{2}{\sqrt{5}}a \right)$$

$$AC' = \frac{\sqrt{85}}{5}a$$

Final Answer : (C)

53) **Explain Question :** Minimum area from intersection of plane with lines.

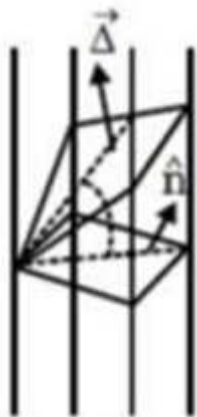
Concept: Projection and symmetry.

Solution :

$$\text{Minimum area} = |\vec{\Delta} \cdot \vec{n}|, \text{ where } \vec{\Delta} = \overline{AB} \times \overline{AC} = -9\vec{j}$$

$$\vec{n} = \frac{1}{3} (2\vec{i} + \vec{j} + 2\vec{k})$$

Required area = 3 square units.



Final Answer : (C)

$$S = \sum_{n=1}^{99} \sqrt{10 + \sqrt{n}} \quad \text{and} \quad T = \sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}$$

54) **Explain Question :** Find ratio of

Concept : Quadratic roots, symmetric radicals.

Solution :

Let $S = \sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}$, $T = \sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}$. Note that $\sqrt{a+b+2\sqrt{ab}} = \sqrt{a} + \sqrt{b}$ for positive real numbers a and b . In particular, for each fixed positive integer n with $1 \leq n \leq 99$, choose $a > b > 0$ satisfying $a + b = 20$, $ab = n$, then a, b are roots of the equations $x^2 - 20x + n = 0$ i.e.

$$a = 10 + \sqrt{100 - n}, \quad b = 10 - \sqrt{100 - n}$$

$$\text{so } \sqrt{2}S = \sum_{n=1}^{99} \sqrt{20 + 2\sqrt{n}} = \sum_{n=1}^{99} \left(\sqrt{10 + \sqrt{100 - n}} + \sqrt{10 - \sqrt{100 - n}} \right)$$

$$= \sum_{n=1}^{99} \left(\sqrt{10 + \sqrt{n}} + \sqrt{10 - \sqrt{n}} \right) = S + T$$

$$\text{Thus, } (\sqrt{2} - 1)S = T \Rightarrow \frac{S}{T} = \frac{1}{\sqrt{2} - 1} = 1 + \sqrt{2}$$

Final Answer : (C)

55)

Explain Question : Sum of two series is given.

Concept : Recurrence relations, telescoping.

Solution :

$$\text{As, } S = \frac{S_1}{2} + \frac{S_2}{2^2} + \dots + \frac{S_9}{2^9} + \frac{S_{10}}{2^{10}}$$

$$\Rightarrow \frac{S}{2} = \frac{b_1}{2^2} + \frac{b_2}{2^3} + \dots + \frac{b_9}{2^{10}} + \frac{b_{10}}{2^{11}}$$

Subtracting

$$\Rightarrow \frac{S}{2} = \frac{b_1}{2} + \left(\frac{a_1}{2} + \frac{a_2}{2^2} \dots + \frac{a_9}{2^9} \right) - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow S = b_1 - \frac{b_{10}}{2^{10}} + \left(\frac{a_1}{2} + \frac{a_2}{2^2} \dots + \frac{a_9}{2^9} \right)$$

$$\Rightarrow \frac{S}{2} = \frac{b_1}{2} - \frac{b_{10}}{2^{11}} + \left(\frac{a_1}{2^2} + \frac{a_2}{2^3} \dots + \frac{a_9}{2^{10}} \right)$$

Subtracting

Put $n = 6$, $\overline{1 - \cos \frac{\pi}{6}} = \overline{2 - \sqrt{3}} = 12$

$$(where\ x\ is\ \frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3})$$

$$\cos(\pi - 2\theta) = \frac{\frac{x^2}{4} + \frac{x^2}{4} - x'^2}{2 \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)}$$

$$\frac{-x^2}{2} \cos 2\theta = \frac{x^2}{2} - x'^2$$

$$x'^2 = \frac{x^2}{2} (1 + \cos 2\theta) = x^2 \cos^2 \theta$$

$$\square \quad x' = x \cos \theta.$$

Final Answer : (D)

57) **Explain Question :** Recursive function telescopes to 4047.

Concept : Fixed points of recurrence.

Solution :

$$a_{i+1} = \frac{-a_i + a_i \sqrt{4a_i - 3}}{2}$$

$$\Rightarrow (a_i)^3 = (a_{i+1})^2 + (a_i)^2 + a_i a_{i+1}$$

Hence

$$\sum_{i=1}^{2023} \frac{(a_{i+1}^3 - a_i^3 + a_i^3)}{a_i^2 + a_{i+1}^2 + a_i a_{i+1}} = \sum_{i=1}^{2023} (a_{i+1} - a_i) + 2023$$

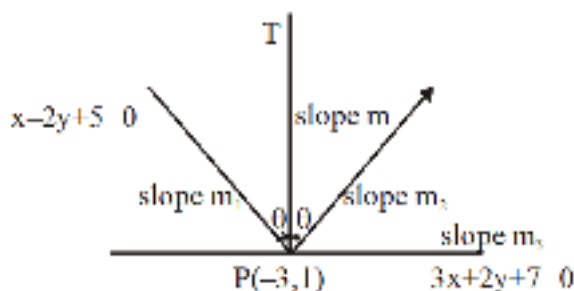
$$= a_{2024} - a_1 + 2023 = 4047 - a_1$$

Final Answer : (B)

58) **Explain Question :** A ray of light along to $x - 2y + 5 = 0$ to the mirror $3x + 2y + 7 = 0$ then equation of reflected ray find

Concept : Reflection using slopes.

Solution :



First we will find the intersection points of these two lines,

$$x - 2y + 5 = 0 \quad \dots(1)$$

$$3x + 2y + 7 = 0 \quad \dots(2)$$

from eqn. (1) & eqn. (2), $x = -3$ & $y = 1$

hence coordinate of P is $(-3, 1)$

To find the eqn. of the reflected ray, we have to find the slope of the reflected ray

$$\text{slope } m_1 = \frac{1}{2}, \quad \text{slope } m_3 = \frac{-3}{2}$$

Line T is perpendicular to $3x + 2y + 7 = 0$

hence, $m \times m_3 = -1$

$$m = \frac{-1}{m_3} = \frac{-1}{-3/2} \times 2 = \frac{2}{3}$$

As angle between the incident ray & line T is same as angle between reflected ray & line T

$$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m}{1 + m_1 m} \right| = \left| \frac{m - m_2}{1 + m m_2} \right| \\&= \left| \frac{\frac{1}{2} - \frac{2}{3}}{1 + \frac{1}{2} \times \frac{2}{3}} \right| = \left| \frac{\frac{2}{3} - m_2}{1 + \frac{2}{3} m_2} \right| \\&\Rightarrow \left| \frac{\frac{3-4}{8}}{\frac{3+2m_2}{3+2m_2}} \right| = \left| \frac{\left(\frac{2-m_2(3)}{3+2m_2} \right)}{\frac{3+2m_2}{3+2m_2}} \right| = \left| \frac{-1}{8} \right| = \left| \frac{2-3m_2}{3+2m_2} \right| \\&\Rightarrow \frac{1}{8} = \left| \frac{2-3m_2}{3+2m_2} \right| \Rightarrow \pm \frac{1}{8} = \frac{2-3m_2}{3+2m_2}\end{aligned}$$

by taking +ve sign by taking -ve sign →

$$\frac{1}{8} = \frac{2-3m_2}{3+2m_2} \quad \quad \quad \frac{-1}{8} = \frac{2-3m_2}{3+2m_2}$$

$$3+2m_2 = 16-24m_2 \quad \quad -3-2m_2 = 16-24m_2$$

$$26m_2 = 13 \quad \quad 22m_2 = 19$$

$$\boxed{m_2 = \frac{1}{2}} \quad \quad \quad \boxed{m_2 = \frac{19}{22}}$$

This is not possible as incident & reflected rays cannot be parallel

So, the eqn of the reflected ray having point $p(-3,1)$ & slope $m_2 = \frac{19}{22}$

$$(y-1) = \frac{19}{22}(x - (-3)), \quad y-1 = \frac{19}{22}(x+3)$$

$$22y - 22 = 19x + 57, \quad 22y = 19x + 79$$

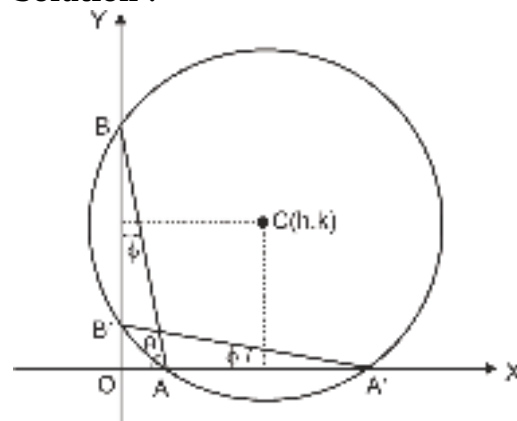
$$\boxed{19x - 22y + 79 = 0}$$

Final Answer : (A)

59) **Explain Question :** A straight line moves upon the axis of X & Y

Concept : Coordinate geometry of rotating rods.

Solution :



Let $\angle OA'B' = \phi$ and $\angle OAB = \theta$

$$\Rightarrow \theta + \phi = \frac{\pi}{2} \text{ and } \angle OBA = \phi$$

∴ length of AB is 'a' and length of A'B' is 'b'

□ from the figure

$$A' (b \cos \phi, 0) \text{ and } A (a \cos \theta, 0)$$

$$\text{similarly } B (0, a \sin \theta) \text{ and } B' (0, b \sin \phi)$$

Let $c(h, k)$ be the centre of circle

$$\square 2h = a \cos \theta + b \cos \phi$$

$$\therefore \theta = \frac{\pi}{2} - \theta$$

$$\square \quad 2h = a \cos \theta + b \sin \theta \quad \dots\dots(i)$$

$$\text{and} \quad 2k = a \sin \theta + b \cos \theta$$

$$\therefore \theta = \frac{\pi}{2} - \theta$$

$$\square \quad 2k = a \sin \theta + b \cos \theta \quad \dots\dots(ii)$$

$$\text{on solving (i) and (ii), we get } \cos \theta = \frac{2ah - 2bk}{a^2 - b^2} \text{ and } \sin \theta = \frac{2ak - 2bh}{a^2 - b^2}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$\square \quad \text{locus of } C(h, k) \text{ is } (2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$$

Final Answer : (D)

60) Question Explanation:

In this question $a + b + c = 0$ is given and based on this condition. We need to determine the nature of a complex algebraic expression involving a, b, c raised to powers up to 6 :

$$a^6 + b^6 + c^6 - \frac{(a^2 + b^2 + c^2)^3}{4} - \frac{(a^3 + b^3 + c^3)^2}{3} \text{ and we have to check the value of this expression is always positive negative or independent of } a, b, c.$$

Concept:

This question is based on Theory of equation & Newton's formula.

Solution:

$$\text{Given } a + b + c = 0$$

$$\text{Let } p = ab + bc + ca \text{ and } q = abc$$

$$\text{Let } S_k = a^k + b^k + c^k$$

$$S_1 = a + b + c = 0$$

$$S_2 = a^2 + b^2 + c^2$$

$$= (a + b + c)^2 - 2(ab + bc + ca)$$

$$= (0)^2 - 2p$$

$$S_2 = -2p$$

$$S_3 = a^3 + b^3 + c^3 \quad (\square \quad a + b + c = 0)$$

$$= 3abc$$

$$S_3 = 3q$$

We can assume a cubic equation whose roots are a, b, c as-

$$\boxed{x^3 + px - q = 0}$$

use Newton's formula.

$$S_{k+3} + pS_{k+1} - qS_k = 0$$

$$S_{k+3} = qS_k - pS_{k+1}$$

$$\mathbf{k = 1}$$

$$S_4 = qS_1 - pS_2$$

$$= q \times 0 - p(-2p)$$

$$S_4 = 2p^2$$

$$\mathbf{k = 2}$$

$$S_5 = qS_2 - pS_3$$

$$= q \times (-2p) - p(3q)$$

$$S_5 = -5pq$$

$$\mathbf{k=3}$$

$$S_6 = qS_3 - pS_4$$

$$= q \times 3q - p(2p^2)$$

$$S_6 = 3q^2 - 2p^3$$

The expression to evaluate

$$E = S_6 - \frac{(S_2)^3}{4} - \frac{(S_3)^2}{3}$$

Is

Put the value of S_2, S_3, S_6 in terms of p and q

$$= 3q^2 - 2p^3 - \frac{(-2p)^3}{4} - \frac{(3q)^2}{3}$$

$$= 3q^2 - 2p^3 + 2p^3 - 3q^2$$

$$\boxed{E = 0}$$

Since the value of this expression is '0'.

So, It is independent of the values of a, b, c .

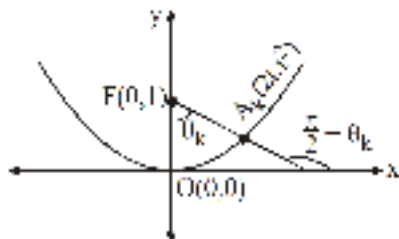
Final Answer:

The correct option is (3)

61) **Explain Question :** Sum of infinite series using definite integration

Concept : Parametric parabola and limit.

Solution :



$$\text{Let } A_k = (2t, t^2)$$

$$\therefore \text{Slope of } FA_k = \frac{t^2 - 1}{2t - 0} = \tan\left(\frac{\pi}{2} + \theta_k\right)$$

$$\therefore \tan \theta_k = \frac{1 - t^2}{2t} = \tan(2\phi) \quad (\text{Say})$$

$$\Rightarrow \phi = \frac{\theta_k}{2} = \frac{k\pi}{4n} \text{ where } \tan \phi = t$$

$$\text{Also } FA_k = \sqrt{(t^2 - 1)^2 + (2t)^2}$$

$$= t^2 + 1 = 1 + \tan^2 \phi = \sec^2\left(\frac{k\pi}{4n}\right)$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n FA_k = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sec^2\left(\frac{\pi}{4} \left(\frac{k}{n}\right)\right)$$

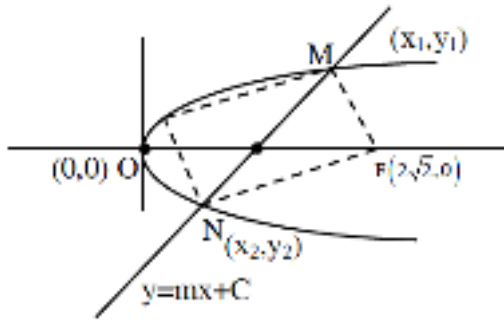
$$= \int_0^1 \sec^2\left(\frac{\pi x}{4}\right) dx = \frac{4}{\pi} \left[\tan \frac{\pi x}{4} \right]_0^1 = \frac{4}{\pi}$$

Final Answer : (B)

62) **Explain Question :** Focal chord of parabola is tangent of hyperbola

Concept : Chord and tangent

Solution :



$$H: \frac{x^2}{4} - \frac{y^2}{4} = 1$$

Focus $(ae, 0)$

$$F(2\sqrt{2}, 0)$$

Line L : $y = mx + c$ pass $(1, 0)$

$$\boxed{0 = m + C} \dots\dots(1)$$

Line L is tangent to Hyperbola. $\frac{x^2}{4} - \frac{y^2}{4} = 1$

$$C = \pm \sqrt{a^2 m^2 - \ell^2}$$

$$C = \pm \sqrt{4m^2 - 4}$$

From (1)

$$-m = \pm \sqrt{4m^2 - 4}$$

Squaring

$$m^2 = 4m^2 - 4$$

$$4 = 3m^2$$

$$\boxed{\frac{2}{\sqrt{3}} = m} \text{ (as } m > 0 \text{)}$$

$$C = -m$$

$$C = \frac{-2}{\sqrt{3}}$$

$$y = \frac{2x}{\sqrt{3}} - \frac{2}{\sqrt{3}}$$

$$y^2 = 4x$$

$$\Rightarrow \left(\frac{2x-2}{\sqrt{3}} \right)^2 = 4x$$

$$\Rightarrow x^2 + 1 - 2x = 3x$$

$$\Rightarrow \boxed{x^2 - 5x + 1 = 0}$$

$$y^2 = 4 \left(\frac{\sqrt{3}y + 2}{2} \right)$$

$$y^2 = 2\sqrt{3}y + 4$$

$$\Rightarrow \boxed{y^2 - 2\sqrt{3}y - 4 = 0}$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 1 & 0 & x_1 & 2\sqrt{2} & x_2 & 0 \\ 2 & 0 & y_1 & 0 & y_2 & 0 \end{vmatrix}$$

$$= \left| \frac{1}{2} [-2\sqrt{2}y_1 + 2\sqrt{2}y_2] \right|$$

$$= \sqrt{2} |y_2 - y_1| = \frac{(\sqrt{2}) \sqrt{12 + 16}}{111}$$

$$= \sqrt{56}$$

$$= 2\sqrt{14}$$

Final Answer : (B)

63) **Explain Question :** Sum of distances from focus to points P,Q,R,S

Concept : Parabola focus property.

Solution :

Any curve passing through point of intersection of $S_1 = x^2 + y^2 + 6x - 24y + 72 = 0$ and

$S_2 = x^2 - y^2 + 6x + 16y - 46 = 0$ is $s_1 + \lambda s_2 = 0$ and its is a parabola

$$\text{So } h^2 = ab$$

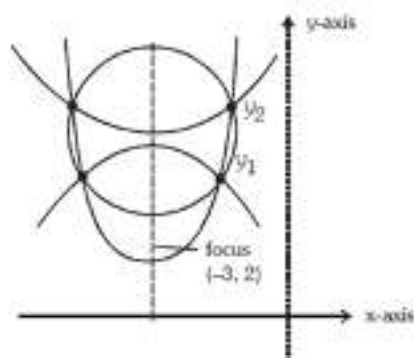
$$\lambda = \pm 1$$

$$\lambda = -1 \text{ (Rejected)}$$

$$\text{For } \lambda = 1$$

$$2x^2 + 12x - 8y + 26 = 0$$

$$(x+3)^2 = 4(y-1)$$



Vertex $(-3, 1)$, Focus $(-3, 2)$

Directrix $y - 1 + 1 = 0$

$$AP + AQ + AS + AR = 2(y_1 + y_2) = 40$$

$$s_1 - s_2 = y^2 - 20y + 59 = 0$$

$$y_1 + y_2 = 20$$

Final Answer : (A)

64) **Explain Question :** Domain analysis yields sum of integers.

Concept : Inverse trigonometric domain.

Solution :

$$f(x) = \frac{\tan^{-1}x}{1 + \frac{2}{\pi}\tan^{-1}x}$$

Clearly,

Now domain of equation $f^2(x) + (\sin^{-1}x)^2 = a$, is $x \in [-1, 1]$

$$\text{Now, } f(x) = \frac{\pi}{2} \left(\frac{\frac{2}{\pi}\tan^{-1}x + 1 - 1}{1 + \frac{2}{\pi}\tan^{-1}x} \right)$$

$$= \frac{\pi}{2} \left(1 - \frac{1}{1 + \frac{2}{\pi}\tan^{-1}x} \right)$$

$$\text{So, } f(x)|_{\min} = \frac{\pi - \pi}{2} \text{ at } x = -1$$

$$f(x)|_{\max} = \frac{\pi}{6} \text{ at } x = 1$$

So, $(f(x))^2 \in \left[0, \frac{\pi^2}{4}\right]$ and minimum and maximum is attained at $x = 0$ and $x = -1$

$$\text{Also, } (\sin^{-1}x)^2 \in \left[0, \frac{\pi^2}{4}\right]$$

$$\therefore a_{\max} = \frac{\pi^2}{2} \text{ at } x = -1 \text{ and } a_{\min} = 0 \text{ at } x = 0$$

$$\text{So, } a \in \left[0, \frac{\pi^2}{2}\right]$$

$$\therefore a_{\text{integer}} = 0, 1, 2, 3, 4$$

Hence, sum of integral values of $a = 10$.

Final Answer : (D)

65) **Explain Question :** Indefinite integration is given

Concept : Integration by substitution method

Solution :

$$I = \int \frac{x^2(1 - \ln x)}{x^4 \left(\left(\frac{\ln x}{x} \right)^4 - 1 \right)} dx$$

$$I = \int \frac{\left(\frac{1 - \ln x}{x^2} \right)}{\left(\left(\frac{\ln x}{x} \right)^4 - 1 \right)} dx$$

$$\text{Put } t = \frac{\ln x}{x}$$

$$\Rightarrow dt = \left(\frac{1 - \ln x}{x^2} \right) dx$$

$$\Rightarrow I = \int \frac{dt}{t^4 - 1} = \int \frac{dt}{(t^2 + 1)(t^2 - 1)} = \frac{1}{2} \int \frac{2dt}{(t^2 + 1)(t^2 - 1)}$$

$$\Rightarrow I = \frac{1}{2} \int \left(\frac{(t^2 + 1) - (t^2 - 1)}{(t^2 + 1)(t^2 - 1)} \right) dt$$

$$\Rightarrow I = \frac{1}{2} \int \left(\frac{1}{t^2 - 1} - \frac{1}{t^2 + 1} \right) dt$$

$$\Rightarrow I = \frac{1}{4} \ln \left| \frac{t-1}{t+1} \right| - \frac{1}{2} \tan^{-1} t + C$$

$$\Rightarrow I = \frac{1}{4} \ln \left| \frac{\left(\frac{\ln x}{x} \right) - 1}{\left(\frac{\ln x}{x} \right) + 1} \right| - \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$$

$$\Rightarrow I = \frac{1}{4} \ln \left| \frac{\ln x - x}{\ln x + x} \right| - \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$$

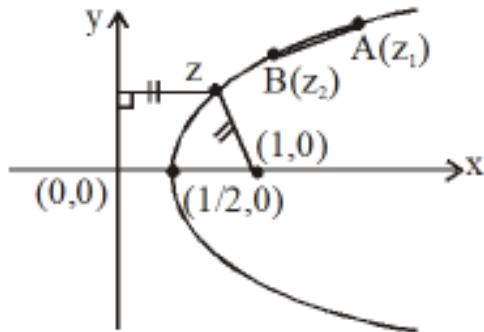
Final Answer : (A)

66) **Explain Question :** Two complex numbers are given, apply given condition.

Concept : Complex locus.

Solution :

$$\begin{aligned} \operatorname{Re}(z) &= \frac{|z-1|}{2} \\ \Rightarrow x &= \sqrt{(x-1)^2 + (y-0)^2} \quad (x > 0) \\ \Rightarrow y^2 &= 2x - 1 = 4 \cdot \frac{1}{2} \left(x - \frac{1}{2}\right) \\ \Rightarrow &\text{a parabola with focus } (1, 0) \text{ \& directrix as imaginary axis.} \\ \square \text{ Vertex} &= \left(\frac{1}{2}, 0\right) \end{aligned}$$



$$\begin{aligned} A(z_1) \text{ \& } B(z_2) \text{ are two points on it such that slope of } AB &= \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \\ (\arg(z_1 - z_2) &= \frac{\pi}{6}) \\ \text{for } y^2 &= 4ax \\ \text{Let } A(at_1^2, 2at_1) \text{ \& } B(at_2^2, 2at_2) \\ m_{AB} &= \frac{2}{t_1 + t_2} = \frac{4a}{y_1 + y_2} = \frac{1}{\sqrt{3}} \\ \left(\text{Here } a = \frac{1}{2} \right) \\ \Rightarrow y_1 + y_2 &= 4a\sqrt{3} = 2\sqrt{3} \\ \text{Final Answer : (D)} \end{aligned}$$

67) **Explain Question :** find the number of elements in the given set.

Concept : Solving Real part equations.

Solution :

$$\begin{aligned} \operatorname{Re}(az^2 + bz) &= a \\ az^2 + bz + a\bar{z}^2 + b\bar{z} &= 2a \\ a(z^2 + \bar{z}^2) + b(z + \bar{z}) &= 2a \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \operatorname{Re}(bz^2 + az) &= b \\ bz^2 + az + b\bar{z}^2 + a\bar{z} &= 2b \\ b(z^2 + \bar{z}^2) + a(z + \bar{z}) &= 2b \quad \dots(2) \end{aligned}$$

$$\begin{aligned} (1) \times b - (2) \times (a) \\ \Rightarrow (b^2 - a^2)(z + \bar{z}) &= 0 \\ \Rightarrow (z + \bar{z}) &= 0 \quad (a^2 \neq b^2) \\ (1) \times a - (2) \times (b) \\ \Rightarrow (a^2 - b^2)(z + \bar{z}) &= 2(a^2 - b^2) \quad (a^2 \neq b^2) \\ z^2 + \bar{z}^2 &= 2 \\ \Rightarrow (z + \bar{z})^2 - 2z\bar{z} &= 2 \\ z\bar{z} &= -1 \\ \Rightarrow 1 + 1^2 &= -1 \\ \Rightarrow &\text{No solution} \end{aligned}$$

But when $a = -b$,

$$\operatorname{Re}(az^2 - az) = a$$

$$\Rightarrow \operatorname{Re}(a(x^2 - y^2 + i2xy) - a(x + iy)) = a$$

$$\Rightarrow a(x^2 - y^2) - ax = a$$

$$\Rightarrow x^2 - y^2 - x = 1$$

$$\Rightarrow x^2 - x - 1 = y^2$$

For any real values of y there two values of x , hence infinite complex numbers are possible.

Final Answer : (C)

68) **Explain Question :** Complex relation is given.

Concept : Parameterization of circle locus.

Solution :

$$z\bar{z} - \sqrt{2}\bar{z}i + \sqrt{2}iz = 0$$

Put $z = x + iy$

$$x^2 + (y - \sqrt{2})^2 = 2 \quad \dots(1)$$

Now

$$f(z) = \frac{6|z|^2 - z^2 - (\bar{z})^2 - 8}{8}$$

Put $z = x + iy$ here

$$= \frac{6(x^2 + y^2) - 2(x^2 - y^2) - 8}{8}$$

$$= \frac{4x^2 + 8y^2}{8} - 1 \quad \dots(2)$$

Since x, y lies on curve (1)

$$x = \sqrt{2} \cos \theta ; y = \sqrt{2} + \sqrt{2} \cos \theta$$

Put in (2)

$$= \frac{40(\sqrt{2} \cos \theta)^2 + 8(\sqrt{2} + \sqrt{2} \sin \theta)^2}{8} - 1$$

$$= 2 + \sin^2 \theta + 4 \sin \theta$$

$$= (\sin \theta + 2)^2 - 2$$

$$m_1 = 7 ; m_2 = -1$$

Final Answer : (C)

69) **Explain Question :** Rotation matrix repeated

Concept : Complex rotation.

Solution :

The point $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n), (a_{n+1}, b_{n+1})$

Given $(a_{n+1} + ib_{n+1}) = \text{say } z_{n+1}$

$$z_{n+1} = (\sqrt{3}a_n - b_n) + i(\sqrt{3}b_n + a_n)$$

$$\sqrt{3}(a_n + ib_n) + i(a_n + ib_n) = \sqrt{3}z_n = \sqrt{3}z_n + iz_n$$

$$= z_n(\sqrt{3} + i) \text{ where } z_n = a_n + ib_n$$

$$= 2z_n \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$\text{Hence, } z_{n+1} = 2z_n e^{i\frac{\pi}{6}}$$

$$\frac{z_{n+1}}{z_n} = 2e^{i\frac{\pi}{6}} \text{ say } \alpha$$

Put, $n = 1, 2, 3, \dots, 99$

$$\frac{z_2}{z_1} = \alpha; \frac{z_3}{z_2} = \alpha, \dots, \frac{z_{100}}{z_{99}} = \alpha$$

Multiplying

$$\frac{z_{100}}{z_1} = \alpha^{99} = 2^{99} e^{i \frac{33\pi}{2}} = 2^{99} i$$

$$z_{100} = 2^{99} (z_1) i$$

$$z_1 = \frac{z_{100}}{2^{99} i} = \frac{(2 + 4i)}{2^{99} i}$$

$$z_1 = 2^{-97} - 2^{-98} i$$

$$a_1 + ib_1 = 2^{-98} i$$

$$a_1 + ib_1 = 2^{-97} - 2^{-98} i$$

$$a_1 = 2^{-97}; b_1 = -2^{-98}$$

$$a_1 + b_1 = 2^{-98} (2 - 1) = 2^{-98}$$

Final Answer : (C)

70) **Explain Question :** Left and right limits differ

Concept : One-sided limits.

Solution :

For $f(0^-)$, let $\sin x = h$

$$\Rightarrow f(0^-) = \lim_{h \rightarrow 0} \frac{h(-1)^2 \ln 2}{\ln(1+h)} = \ln 2$$

for $f(0^+)$ let $\tan x = h$

$$\Rightarrow f(0^+) = \lim_{h \rightarrow 0} \frac{\ln(e^{h^2} + 2\sqrt{h})}{\tan \sqrt{h}}$$

$$\lim_{h \rightarrow 0} \frac{e^{h^2} - 1 + 2\sqrt{h}}{\tan \sqrt{h}} \Rightarrow f(0^+) = 2$$

Final Answer : (D)

71) **Explain Question :** Signum non-diff where argument=0; discriminants analysis

Concept : Roots of polynomial.

Solution :

$$f(x) = \text{sgn} \left((x^2 - ax + 1) (bx^2 - 2bx + 1) \right)$$

For exactly one point of discontinuity $(x^2 - ax + 1) (bx^2 - 2bx + 1) = 0$ at exactly one value of x .

Let $P(x) = x^2 - ax + 1$ (1) and $Q(x) = bx^2 - 2bx + 1$ (2) $b \neq 0$

and D_1 and D_2 are the discriminant of equation (1) and (2) respectively.

Case-I: $D_1 = 0$ and $D_2 < 0$

$$a^2 - 4 = 0 \quad 4b^2 - 4b < 0$$

$$a = \pm 2 \quad b \in (0, 1) \text{ rejected because there is no integral value of } b.$$

Case-II: $D_1 < 0$ and $D_2 = 0$

$$a \in (-2, 2) \quad b = 0, 1$$

$$a = -1, 0, 1 \quad b = 0 \text{ is rejected}$$

□ number of ordered pairs are 3.

Case-III: $D_1 = 0$ and $D_2 = 0$

$$a = -2, 2 \quad b = 0, 1$$

For $b = 0$, $a = -2$ & 2 and for $b = 1$, $a = 2$

- number of ordered pairs are 3.
- Total number of ordered pairs are 6

Final Answer : (6)

72) **Explain Question :** Continuity at $\pi/2$

Concept : One-sided limits via substitution.

Solution :

Hence $f(x)$ is continuous at $x = \frac{\pi}{2}$

$$\Rightarrow K = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{(\pi - 2x)^2} \cdot \frac{\log \sin x}{\log(1 + \pi^2 - 4\pi x + 4x^2)}$$

Put $\alpha = \left(\frac{\pi}{2} - x\right)$

$$K = \lim_{\alpha \rightarrow 0} \frac{1 - \cos \alpha}{(2\alpha)^2} \cdot \frac{\log \cos \alpha}{\log(1 + 4\alpha^2)}$$

$$= \lim_{\alpha \rightarrow 0} (1 - \cos \alpha) \frac{\log(1 + \cos \alpha - 1)}{(\cos \alpha - 1)} \frac{4\alpha^2}{\log(1 + 4\alpha^2)} \frac{(\cos \alpha - 1)}{(2\alpha)^2 4\alpha^2}$$

$$= \lim_{\alpha \rightarrow 0} \frac{1 - \cos \alpha}{\left(\frac{\alpha^2}{4}\right)} \frac{(\cos \alpha - 1)}{\frac{\alpha^2}{4}} \frac{\frac{\alpha^2}{4} \cdot \frac{\alpha^2}{4}}{4\alpha^2 \cdot 4\alpha^2}$$

$$K = \frac{2 \cdot (-2)}{4 \cdot 4 \cdot 4 \cdot 4} = -\frac{1}{64}$$

Final Answer : (64)

73) **Explain Question :** Continuity gives quadratic with roots 2,3

Concept : Root conditions.

Solution :

$$\because \frac{\pi}{3} < A < \frac{\pi}{2} \Rightarrow \frac{\sqrt{3}}{2} < \sin A < 1 \text{ and } \frac{3\sqrt{3}}{8} < \sin^3 A < 1$$

$$0 < \sin A - \sin^3 A < 1 \text{ and } \sin A + \sin^3 A > 1$$

$$\therefore f(x) = \begin{cases} ax^2, & x \in \mathbb{Q} \\ 5x - b, & x \notin \mathbb{Q} \end{cases}$$

$\therefore f(x)$ is continuous at $x = 2$ and 3

$$\therefore ax^2 = 5x - b \text{ should have roots } 2 \text{ and } 3 \Rightarrow ax^2 - 5x + b = 0$$

$$\therefore \frac{5}{a} = 5 \Rightarrow a = 1 \text{ and } \frac{b}{a} = 6 \Rightarrow b = 6$$

$$\therefore b - a = 5$$

Final Answer : (5)

74) **Explain Question :** Circle tangent + normal to parabola

Concept : Normal and tangent geometry.

Solution :

Equation of normal at point $T\left(t, \frac{t^2}{2}\right)$ to C is $y - \frac{t^2}{2} = \frac{-1}{t}(x - t) \Rightarrow y = \frac{-x}{t} + \frac{t^2}{2} + 1$

If circle S touches curve C at T.

$\Rightarrow (x(t), y(t))$ must lie on normal

$$\Rightarrow y(t) = \frac{-x(t)}{t} + \frac{t^2}{2} + 1$$

Also distance of centre of circle from T equals radius = distance of centre of circle from x-axis.

$$\Rightarrow (x(t) - t)^2 + \left(y(t) - \frac{t^2}{2}\right)^2 = (y(t))^2$$

$$x(t) = \frac{t(1 + \sqrt{t^2 + 1})}{2}, y(t) = \frac{t^2 + 1 - \sqrt{t^2 + 1}}{2}$$

$$\Rightarrow x(t) y(t) = \frac{1}{4} t^3 \sqrt{t^2 + 1}$$

$$\lim_{r \rightarrow 0^+} \int_r^1 \frac{1}{4} t^3 \sqrt{t^2 + 1} dt$$

$$\text{Let } \sqrt{1 + t^2} = z \Rightarrow 2tdt = 2zdz$$

$$\Rightarrow \lim_{r \rightarrow 0^+} \int_{\sqrt{1+r^2}}^{\sqrt{2}} \frac{1}{4} (z^2 - 1) z^2 dz = \frac{\sqrt{2} + 1}{30}$$

Final Answer : (5)

75) **Explain Question :** Limit using f(x) & g(x) conditions

Concept : Derivative limits.

Solution :

$$g(x) = f(x) \cdot t \rightarrow g'(x) = f'(x) \cdot t + f(x) \frac{dt}{dx}$$

$$\frac{g'(x)}{f'(x)} = \frac{g(x) - \sqrt{f^2(x) + g^2(x)}}{f(x)} \rightarrow \frac{g(x)}{f(x)} = t$$

$$\Rightarrow \ln(\sqrt{1+t^2} + t) + \ln f(x) = \ln c$$

$$\Rightarrow \sqrt{g^2(x) + f^2(x)} + g(x) = C$$

$$f(0) = 10, g(0) = 0 \Rightarrow C = 10$$

$$\lim_{x \rightarrow \infty} \sqrt{g^2(x) + f^2(x)} + \lim_{x \rightarrow \infty} g(x) = 10$$

$$\lim_{x \rightarrow \infty} g(x) = 5$$

Final Answer : (5)